

Continuum ED and symmetry-projected neural Bloch states of twisted MoTe₂

Local benchmarks for CDW and fractional Chern-insulator calculations

Local numerical benchmark for the Physics-for-AI/QNS programme

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Abstract

We establish a symmetry-resolved exact-diagonalization (ED) benchmark and a continuous-coordinate neural-Bloch experiment for twisted MoTe₂. The continuum calculation reproduces the one-five-band reference energies, while one-band 27-cell ED distinguishes the $\nu = 1/3$ charge-density-wave tower from the approximately three-fold Γ manifold at $\nu = 2/3$. A tracked three-state calculation gives Chern numbers near zero and two, but complete all-sector spectral isolation, large-gauge sewing and twist-grid convergence remain required before assigning a quantized Hall response.

The variational calculation follows the Luo-Fu Appendix-B neural-orbital construction in the full plane-wave continuum basis. Version 4 retains one dense trainable determinant matrix M and applies the exact finite-translation projector to the complete generalized determinant. By Cauchy-Binet this contains all and only total- Γ minors, without explicitly enumerating momentum combinations or adding a second mixing tensor. A subsequent complete-wavefunction C_3 projector resolves $m = 0, 1, 2$. The three branches use independent parameters with 20 shared-mixture-sampling updates followed by 100 private-chain updates. Following the reference convention, reported energies are final-ten means of the persistent training chain; their RMS is a tail-stability diagnostic rather than an independent error bar. At nine cells, the lowest outer branches are $m = 0$ and $m = 2$ at fillings $1/3$ and $2/3$, with final-ten energies $-37.3215(244)$ and $-52.9124(616)$ meV per particle. The width-64, 27-cell extension instead gives $-47.5342(347)$ meV in $m = 0$ at $1/3$ and $-60.4155(391)$ meV in $m = 1$ at $2/3$. These are single-seed correlated optimization traces in the full continuum Hamiltonian, not controlled bounds against band-truncated ED. Figures 6–10 compare nine- and 27-cell energies, timings and observables. Their two $n(\mathbf{k})$ estimators are the Bloch-projected one-body-density-matrix diagonal summed over bands 1–5 and its first-band component, the latter conditionally normalized to $\sum_{\mathbf{k}} n_1(\mathbf{k}) = N_e$ for one-band ED comparison. The full $S(\mathbf{q})$ is the unbinned Fourier transform of the sampled real-space density-pair correlation at exactly the 27 physical ED momenta. A separately evaluated 37-vector,

C_6 -closed grid audits the hexagonal-BZ boundary without replacing $\mathbf{q}$ by $\mathbf{q} + \mathbf{G}$. No observable is C_3 averaged or fit to the ED momentum profile [1, 3]. Observables use the saved 4,128-walker training ensembles without a validation chain. Figure 11 combines the nine/27-cell Krylov stability audit.

Part I Exact-diagonalization benchmark

1 Motivation

Fractional Chern insulators (FCIs) in moiré semiconductors require simultaneous control of strong Coulomb correlations, band mixing and nonuniform Bloch-state quantum geometry. A quantum neural state (QNS) can reach larger systems and avoid a rigid few-band truncation, but a lower variational energy is not sufficient evidence of the correct phase. The reference calculation must also reproduce symmetry quantum numbers, the torus ground-state counting, momentum occupation and density correlations.

The present benchmark therefore separates two questions. First, the 3×3 multiband calculation tests the continuum Hamiltonian and the variational convergence from one to five retained bands, directly against Fig. 2 of Ref. [1]. Second, the one-band 27-site calculation repeats the phase diagnostics of Supplementary Fig. 5: a Γ, K, K' CDW tower at $\nu = 1/3$ versus an approximately three-fold Γ manifold at $\nu = 2/3$ [1]. The latter momentum counting is supplemented below by flux spectral flow and a gauge-invariant many-body Chern number.

2 Continuum model and quantum geometry

For a fixed spin/valley sector, the single-hole Hamiltonian is

$$H_0 = \begin{pmatrix} \frac{\hbar^2(-i\nabla - \boldsymbol{\kappa}_+)^2}{2m^*} + V_1(\mathbf{r}) & t(\mathbf{r}) \\ t^*(\mathbf{r}) & \frac{\hbar^2(-i\nabla - \boldsymbol{\kappa}_-)^2}{2m^*} + V_2(\mathbf{r}) \end{pmatrix}. \quad (1)$$

The lowest moiré harmonics and plane-wave convention follow Refs. [6, 1]. We use

$$(V, w, \phi, m^*) = (11.2 \text{ meV}, 13.3 \text{ meV}, -91^\circ, 0.62m_e), \quad (2)$$

monolayer lattice constant $a_0 = 0.352 \text{ nm}$, and eight radial reciprocal shells. The resulting Bloch Hamiltonian has 61 plane waves per layer. Figure 1 compares the first ten moiré bands at $\theta = 2.6^\circ$ and 3.0° .

For a nondegenerate band n , the quantum geometric tensor is evaluated from velocity matrix elements,

$$Q_{\alpha\beta}^{(n)}(\mathbf{k}) = \sum_{m \neq n} \frac{\langle n | v_\alpha | m \rangle \langle m | v_\beta | n \rangle}{[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2}, \quad (3)$$

with $g_{\alpha\beta} = \text{Re } Q_{\alpha\beta}$ and $\Omega = -2 \text{Im } Q_{xy}$. Bands 1–3 are calculated on a 49-point linear grid in each reciprocal direction, restricted to the hexagonal Brillouin zone, and smoothly triangulated only for visualization. The shared colorbars retain physical one-dimensional numerical ticks. Values outside the 1.5–98.5 percentile interval are clipped in the rendering, not in the saved data.

3 Projected interaction and ED symmetries

The periodically continued Coulomb interaction is projected into N_b Bloch bands,

$$H = \sum_{n\mathbf{k}} \varepsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}} + \sum_{ab,cd} V_{cd,ab} c_c^\dagger c_d^\dagger c_b c_a, \quad (4)$$

where $V(\mathbf{q}) = 2\pi e^2/(\epsilon|\mathbf{q}|)$ and the uniform $\mathbf{q} = 0$ component is removed by the neutralizing background. Eight radial reciprocal shells are retained in the form-factor sum.

Charge- $U(1)$ is enforced by fixed hole number, while spin- $U(1)$ is enforced by restricting to the fully polarized sector. Translation symmetry first decomposes the Hilbert space into many-body momentum blocks $\mathcal{H}_K$. At generic momentum, a rotation maps $\mathcal{H}_K$ to the distinct block $\mathcal{H}_{C_n K}$, so point-group symmetry is used only to copy spectra within a momentum orbit. A further C_3/C_6 character decomposition is available only when $C_n K = K + G$, i.e. when the rotation belongs to the little group of K .

The phase-diagnostic cluster has real-space supercell matrix

$$S_{27} = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad \det S_{27} = 27, \quad (5)$$

and is therefore the $3\sqrt{3} \times 3\sqrt{3}$ C6 torus rather than a rectangular 3×9 mesh. Its momentum group is written as $\mathbb{Z}_3 \times \mathbb{Z}_9$, with plot index

$$I_k = k_1 + N_1 k_2, \quad N_1 = 3, \quad I_k = 0, \dots, 26. \quad (6)$$

In this convention Γ, K', K occur at $I_k = 0, 9, 18$. The one-band Hilbert space contains $\binom{27}{9} = 4,686,825$ states at $\nu = 1/3$. At $\nu = 2/3$, an exactly verified particle–hole transformation maps 18 particles to 9 holes without changing the spectrum.

4 Multiband benchmark on the 3×3 cluster

Figure 2 summarizes the 3×3 calculation at $\theta = 3^\circ$, $\epsilon = 5$. The ground-state energy decreases monotonically as the retained band space grows. The values in Table 1 reproduce the available $\nu = 2/3$ points in Fig. 2 of Ref. [1] within 0.036 meV per hole. Remote-band mixing is visibly stronger at $\nu = 2/3$.

Table 1: Ground-state energy per hole in meV at $\theta = 3^\circ$, $\epsilon = 5$ on the 3×3 cluster.

N_b	$\nu = 1/3$	$\nu = 2/3$	paper Fig. 2
1	−36.3296	−47.1561	−47.15
2	−37.1158	−50.0774	−50.06
3	−37.3518	−51.8605	−51.83
4	−37.3878	−52.5379	—
5	−37.3932	−52.7255	−52.69

The middle column of Fig. 2 shows the lowest energy in every momentum sector for one-, three- and five-band ED. Energies are plotted as $E_K - E_{\text{GS}}$ in meV, without division by particle number. The right column reports the additive band-resolved weight

$$w_b = \frac{\langle N_b \rangle}{N_h}, \quad \sum_{b=1}^{N_b} w_b = 1. \quad (7)$$

For five bands, the weights are

$$\nu = 1/3 : (0.96469, 0.02953, 0.00523, 0.00049, 0.00005), \quad (8)$$

$$\nu = 2/3 : (0.88952, 0.07591, 0.02535, 0.00763, 0.00159). \quad (9)$$

These are normalized band populations, not overlaps between separately optimized truncated Hilbert spaces.

5 Reproduction of Supplementary Fig. 5

Figure 3 repeats the one-band 27-site benchmark at $\theta = 2.6^\circ$, $\epsilon = 5$, and adds the ground-manifold averaged $n(\mathbf{k})$.

At $\nu = 1/3$, the three global low states occur at Γ, K, K' with

$$(E_\Gamma, E_K, E_{K'}) - E_{\text{GS}} = (0, 0.573, 0.573) \text{ meV}. \quad (10)$$

The next state is 1.476 meV above the top of this manifold. The K, K' states fold to Γ in the Brillouin zone of the tripled CDW unit cell, matching the interpretation of Ref. [1].

At $\nu = 2/3$, the three lowest states all lie in the Γ sector,

$$(E_{\Gamma,0}, E_{\Gamma,1}, E_{\Gamma,2}) - E_{\text{GS}} = (0, 0.108, 0.298) \text{ meV}. \quad (11)$$

Thus the finite-size threefold splitting is small but explicitly nonzero. The gap between the highest manifold state and the next global level is 3.482 meV. This reproduces the torus counting required for the $\nu = 2/3$ FCI/FQAH state.

The absolute energy origins used to construct the two spectra in Fig. 3 are listed explicitly in Table 2. They are the global minima over all 27 translation sectors; the plotted ordinate is obtained by subtracting the corresponding total energy in the fourth column, without dividing the excitation energy by particle number.

Table 2: Absolute one-band, 27-cell ground-state energies used as the zero of the Fig. 3 spectra.

ν	N_h	three-state manifold	E_{GS} (meV)	E_{GS}/N_h (meV)
1/3	9	Γ, K, K'	-428.359163	-47.595463
2/3	18	$\Gamma_0, \Gamma_1, \Gamma_2$	-1034.203144	-57.455730

For each filling, $n(\mathbf{k})$ and the projected static structure factor are averaged over the three-state ground manifold,

$$n(\mathbf{k}) = \langle c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \rangle, \quad (12)$$

$$S(\mathbf{q}) = \frac{1}{N_h} \langle \rho(-\mathbf{q}) \rho(\mathbf{q}) \rangle, \quad \mathbf{q} \neq 0. \quad (13)$$

The connected value $S(0)$ is set to zero. The $n(\mathbf{k})$ sums are exactly 9 and 18. For Fig. 3, the discrete $n(\mathbf{k})$ markers are placed over an interpolated, weak band-1 Berry-curvature background. The $S(\mathbf{q})$ markers in both rows are placed over the same interpolated band-1 trace-condition deviation $\text{Tr } g_1(\mathbf{q}) - |\Omega_1(\mathbf{q})|$. The background palettes match Fig. 1, while the observable markers use brighter independent color scales.

At $\nu = 1/3$, $S(K) = S(K') = 1.057$, producing the Bragg peaks characteristic of CDW order. At $\nu = 2/3$, the maximum at the same momenta is only 0.197 and is comparable to the surrounding finite- q response; no CDW Bragg peak is present. Together with the three Γ states, this is consistent with the FQAH liquid signature reported in Supplementary Fig. 5.

6 Twisted-boundary spectra and many-body Berry curvature

Let $\mathbf{a}_1, \mathbf{a}_2$ be the primitive moiré real-space vectors and let $\mathbf{b}_a \cdot \mathbf{a}_b = 2\pi \delta_{ab}$. The C_6 -compatible 27-cell ED torus is

generated by the columns of

$$S = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad |\det S| = 27, \quad (14)$$

$$\mathbf{L}_1 = 3\mathbf{a}_1 + 3\mathbf{a}_2,$$

$$\mathbf{L}_2 = -3\mathbf{a}_1 + 6\mathbf{a}_2.$$

For every particle i , the many-body twisted boundary condition is

$$\Psi(\dots, \mathbf{r}_i + \mathbf{L}_a, \dots) = e^{i\phi_a} \Psi(\dots, \mathbf{r}_i, \dots), \quad \phi_a \in [0, 2\pi]. \quad (15)$$

Writing a one-particle momentum as $\mathbf{k} = B\mathbf{f}$, with $B = (\mathbf{b}_1, \mathbf{b}_2)$, Eq. (15) gives $S^T \mathbf{f} = \mathbf{m} + \phi/(2\pi)$. Hence the 27 allowed momenta are

$$\mathbf{f}_{\mathbf{m}}(\phi) = S^{-T} \left(\mathbf{m} + \frac{\phi}{2\pi} \right), \quad \mathbf{m} \in \mathbb{Z}^2 / S^T \mathbb{Z}^2, \quad (16)$$

$$\mathbf{k}_{\mathbf{m}}(\phi) = \mathbf{k}_{\mathbf{m}}(0) + \delta \mathbf{k}, \quad \delta \mathbf{k} = BS^{-T} \frac{\phi}{2\pi}. \quad (17)$$

The code enumerates this quotient group with exact integer residues, so precisely 27 inequivalent momenta are retained and every two-body momentum addition is performed modulo $S^T \mathbb{Z}^2$. The torus is therefore implemented periodically without rounding Cartesian momenta.

At every (ϕ_1, ϕ_2) , the continuum Hamiltonian is solved anew at all $\mathbf{k}_{\mathbf{m}}(\phi)$, after which the one-band Coulomb form factors and projected interaction are rebuilt from the shifted Bloch spinors. A projected scattering term is retained only when $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4$ in the finite translation group (equivalently up to moiré and supercell reciprocal vectors). Charge- $U(1)$ and translations remain exact. A generic twist breaks C_3/C_6 , so rotational orbit copying is not used away from little-group-invariant twists. At $\nu = 2/3$, the 18-electron Hamiltonian is evaluated exactly through its nine-hole particle-hole transform, including its constant and induced one-body terms; this leaves the boundary condition and spectrum unchanged.

Because the three lowest states are quasi-degenerate, an individual-eigenvector Berry phase is gauge dependent. We instead use the determinant link of the full three-state subspace,

$$M_\mu^{ab}(\phi) = \langle \Psi_a(\phi) | \Psi_b(\phi + \Delta_\mu) \rangle, \quad (18)$$

$$U_\mu(\phi) = \frac{\det M_\mu(\phi)}{|\det M_\mu(\phi)|}, \quad (19)$$

$$\Phi_{12}(\phi) = \arg[U_1(\phi)U_2(\phi + \Delta_1)U_1^*(\phi + \Delta_2)U_2^*(\phi)]. \quad (20)$$

The overlap in M_μ is not a bare dot product of ED coefficient vectors, because the one-body Bloch basis itself changes with twist. For an electron configuration I , its coefficient overlap is weighted by

$$w_I = \prod_{p \in I} \langle u_p(\phi) | u_p(\phi + \Delta_\mu) \rangle. \quad (21)$$

For the nine-hole representation, the code uses the exactly equivalent electron complement,

$$w_H = \frac{\prod_{p=1}^{27} \langle u_p(\phi) | u_p(\phi + \Delta_\mu) \rangle}{\prod_{h \in H} \langle u_h(\phi) | u_h(\phi + \Delta_\mu) \rangle}. \quad (22)$$

The non-Abelian FHS curvature and discrete Chern number are

$$\begin{aligned}\mathcal{F}_{\phi_1\phi_2} &= \frac{\Phi_{12}}{(\Delta\phi)^2}, \\ C_{\text{bundle}} &= \frac{1}{2\pi} \sum_{i,j} \Phi_{12}(\phi_{ij}), \\ \Delta\phi &= \frac{2\pi}{6}.\end{aligned}\tag{23}$$

We solve and cache the 7×7 grid endpoints as well as the 6×6 plaquettes. Links crossing a 2π boundary are consequently evaluated from endpoint Bloch overlaps, rather than by identifying raw ED eigenvectors. A separate analytic large-gauge sewing-matrix test is nevertheless still needed for a formal convergence claim. This single grid is a discrete diagnostic, not a twist-grid convergence study.

Figure 4 shows the two axial flux spectra and the curvature map. For $\nu = 1/3$, the bundle contains the lowest states from Γ, K, K' . Its Chern number is

$$C_{\text{bundle}}^{1/3} = 9.3 \times 10^{-7} \simeq 0,\tag{24}$$

and the smallest sampled separation between the three-state manifold and the nearest state in these three sectors is 1.476 meV. For $\nu = 2/3$, the three states all belong to Γ and

$$C_{\text{tracked}}^{2/3} = 1.99999975 \simeq 2.\tag{25}$$

If this tracked subspace is subsequently shown to be the globally isolated ground-state bundle, the conditional Hall response is $\sigma_{xy} = C_{\text{tracked}} e^2 / (3h) = 2e^2 / (3h)$. The minimum same- Γ separation is 5.144 meV, the minimum link singular value is 0.958 at $\nu = 2/3$, and the minimum tracked-sector overlap is 0.964 at $\nu = 1/3$. These checks show that the sampled links are nonsingular, but they do not prove isolation from the complete spectrum. Missing checks are the lowest relevant levels in all 27 translation sectors at every twist, the global $\min_{\phi}(E_4 - E_3)$, an explicit analytic 2π large-gauge sewing test, and $6 \times 6 \rightarrow 9 \times 9 \rightarrow 12 \times 12$ grid convergence.

7 ED benchmark protocol for QNS

The validated workflow is:

1. solve the plane-wave continuum Hamiltonian and verify shell convergence;
2. calculate band dispersion, Berry curvature and quantum metric on the same Bloch convention;
3. project the neutral-background Coulomb interaction with reciprocal-vector form factors;
4. fix charge- $U(1)$ and spin- $U(1)$, then block by translation; use point-group orbit equivalence at generic K , and split by C_3/C_6 character only when the rotation belongs to the little group of K ;

5. use sparse Hermitian Lanczos independently in each required sector;
6. compare energies, sector counting, normalized band weights, $n(\mathbf{k})$ and $S(\mathbf{q})$;
7. insert boundary flux, track the isolated ground manifold and integrate its non-Abelian Berry curvature;
8. only then benchmark the QNS energy, variance and state overlaps against ED.

Numerical regression checks support every stage. The general C6-supercell interaction agrees with the earlier rectangular 3×3 implementation to relative error 3.3×10^{-16} . Direct electron and particle-hole spectra agree sector by sector to 3.4×10^{-13} meV. The finite plane-wave cutoff produces a maximum one-body C6 mismatch of 8.0×10^{-6} meV, and the pre-symmetrization interaction Hermiticity error is 7.8×10^{-9} . The current complete 51-test regression suite passes; the topology outputs additionally record link conditioning, Lanczos residuals and interaction Hermiticity at every twist.

7.1 Wall-clock baseline for comparison with QNS

Table 3 records the measured CPU wall time of the present run. The VMware guest exposed 64 vCPUs and 251 GiB of memory. The sequential stages used 12 OpenMP threads, while the two Berry calculations ran concurrently on disjoint 24-vCPU affinity sets. No GPU was used. Setup and observable stages are identified separately because they should not be confused with Lanczos diagonalization when comparing against a future QNS calculation.

The complete sequential Fig. 2–3 production window was 40 min 48 s. The $\nu = 2/3$ and $\nu = 1/3$ Berry jobs took approximately 26 min 32 s and 34 min 20 s of active wall time, respectively, so their parallel critical path was 34 min 22 s. Future QNS comparisons should additionally record accelerator allocation, peak memory, compilation or warm-up policy, and final energy error relative to this ED baseline.

Part II Neural many-body Bloch wavefunctions

8 Motivation and scope

The second calculation tests the neural many-body Bloch construction of Ref. [1] directly in the continuum model benchmarked in Part I. The purpose is not to fit coefficients in the one-band ED Hilbert space. Instead, a graph network dresses continuum Bloch orbitals in coordinate space and the resulting generalized Slater determinant is optimized against the complete interacting

Table 3: Measured ED wall-clock baseline on the current VMware guest. Berry times exclude the 20 min 6 s reboot interval.

Calculation	Scope	Wall time
Fig. 2, 1–3 bands	3×3 , $\nu = 2/3$, all momentum sectors	45.2 s
Fig. 2, 4 bands	Γ sector, cached operator	15.6 s
Fig. 2, 5 bands	Γ sector, dimension 905,100	11 min 0.5 s
Fig. 2, 5-band spectrum	cached Γ plus two inequivalent sectors	19 min 12.1 s
Fig. 3, one band	27 cells, both fillings, momentum spectra	7 min 21.9 s
Fig. 3 observables	ground manifolds, $n(\mathbf{k})$ and $S(\mathbf{q})$	1 min 3.7 s
Fig. 4 Berry, parallel pair	6×6 plaquettes, 7×7 endpoints	~ 34 min 22 s

continuum Hamiltonian. This can generate remote-band content even though the fixed one-body input is the mean-field first band.

This revision follows Appendix B, Eqs. (16)–(23), of Ref. [1]. In particular, the orbital transformation is the direct complex output

$$J_{\mathbf{k}i}(X) = \text{MLP}([\mathbf{k}, V_i^t]) \in \mathbb{C}, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i} \varphi_{\mathbf{k}i}. \quad (26)$$

There is no constant-one skip, residual parameterization, or factor 0.2; the implemented transformation is the direct $J_{\mathbf{k}i}$ above. The current source additionally implements three audit corrections: full- M uses an unbiased complex random initialization; the node aggregation is the literal $\sum_j m_{ij}$ in Eq. (19), including $j = i$; and the random auxiliary features h_i^0, h_{ij}^0 are explicitly retained in the trainable parameter tree. Version 4 retains one dense complex matrix M but projects the complete generalized determinant onto total momentum Γ by an exact finite translation average. By Cauchy–Binet this keeps all and only the Γ -sector minors, with mutually constrained coefficients $\det M_{:,I}$, without enumerating momentum combinations or introducing a second determinant-mixing tensor. Earlier one-hot- M , no-self-message, single-minor and v3 M_S checkpoints are excluded from the v4 comparison.

Both v4 Γ -projected runs and all six v4 outer- C_3 branches use the same 120-update protocol described below. The internal-equivariance construction investigated in v2 is omitted from the present results because its three-fold evaluation was expensive and did not improve the controlled energy comparison. Quoted training energies use the final ten persistent-chain measurements; the corresponding RMS is a stability diagnostic rather than an independent error bar.

9 Configuration, fixed Bloch basis, and conventions

A walker is a first-quantized configuration

$$X = \{(\mathbf{r}_i, l_i)\}_{i=1}^{N_e}, \quad \mathbf{r}_i \in [0, 1)^2, \quad l_i^{\text{code}} \in \{0, 1\}. \quad (27)$$

The fractional coordinate is converted to a Cartesian coordinate on the 3×3 torus. The layer input used by the network is

$$s_i = 2l_i^{\text{code}} - 1 \in \{-1, +1\}, \quad (28)$$

which is the $l_i = \pm 1$ convention of Ref. [1]. Particle labels are only row labels of the determinant and graph; no ordered occupation string is used.

The nine mesh momenta and plane-wave orbitals are

$$\mathbf{k}_{m_1 m_2} = \frac{m_1}{3} \mathbf{b}_1 + \frac{m_2}{3} \mathbf{b}_2, \quad m_1, m_2 = 0, 1, 2, \quad (29)$$

$$\phi_{\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{l\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}. \quad (30)$$

The reciprocal set contains the same 61 $\mathbf{G}$ vectors per layer used for the single-particle bands in Fig. 1. The coefficients $u_{l\mathbf{k}\mathbf{G}}^l$ are obtained once by diagonalizing the continuum mean-field Hamiltonian. In JAX they are constants, and in PyTorch they are registered buffers. They are absent from the trainable parameter tree. Thus every run uses band 1 as its mean-field orbital source; neither ED amplitudes nor trainable plane-wave coefficients are used.

10 Appendix-B graph network

10.1 Initial physical and auxiliary features

The three reciprocal harmonics used at each vertex are $\mathbf{g}_a \in \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_1 + \mathbf{b}_2\}$. The immutable node feature is

$$\mathbf{v}_i^0 = [\sin(\mathbf{g}_a \cdot \mathbf{r}_i), \cos(\mathbf{g}_a \cdot \mathbf{r}_i), s_i] \in \mathbb{R}^7. \quad (31)$$

For every ordered pair (i, j) , the immutable edge feature is

$$\mathbf{e}_{ij}^0 = [\sin(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), \cos(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), s_i] \in \mathbb{R}^{19}, \quad (32)$$

where all nine mesh momenta are included. In particular, the edge layer channel is the source layer s_i , exactly as in Eq. (16) of Ref. [1]; no additional layer channels are appended.

The complete initial features are

$$\mathbf{V}_i^0 = (v_i^0, h_i^0) \in \mathbb{R}^{32}, \quad \mathbf{E}_{ij}^0 = (e_{ij}^0, h_{ij}^0) \in \mathbb{R}^{32}. \quad (33)$$

The code therefore uses 25 auxiliary node components and 13 auxiliary edge components. Appendix B explicitly calls h_i^0 and h_{ij}^0 “randomly initialized parameters.” They are therefore leaves of the trainable JAX pytree (and

`nn.Parameter` objects in PyTorch), so natural-gradient updates act on them together with the other network parameters. The present code uses one random vector shared over particle labels and one shared over ordered pairs; sharing preserves permutation equivariance. Each component is initialized from $\mathcal{N}(0, 1/32)$. The paper does not specify this distribution or sharing rule, so both remain explicit local conventions. There is no learned input embedding before Eq. (36).

10.2 Two message-passing iterations

For iteration $a = 1, 2$,

$$Q_{ij}^a = W_Q^a E_{ij}^{a-1}, \quad (34)$$

$$K_{ij}^a = W_K^a E_{ij}^{a-1}, \quad (35)$$

$$m_{ij}^a = F^a \left(\sum_n Q_{in}^a \otimes K_{nj}^a \right) \odot G^a(E_{ij}^{a-1}), \quad (36)$$

$$h_i^a = h_i^{a-1} + U^a \left(\left[\sum_j m_{ij}^a, V_i^{a-1} \right] \right), \quad (37)$$

$$h_{ij}^a = h_{ij}^{a-1} + H^a([m_{ij}^a, E_{ij}^{a-1}]), \quad (38)$$

$$V_i^a = (v_i^0, h_i^a), \quad (39)$$

$$E_{ij}^a = (e_{ij}^0, h_{ij}^a). \quad (40)$$

The outer product in Eq. (36) is flattened before F^a . F^a, G^a, H^a, U^a are one-layer real affine maps followed by exact GELU. The two iterations have independent parameters. No division by N_e , no neighbor-count normalization, and no LayerNorm is applied. Self edges remain available to both the edge update and node aggregation. This is the literal $\sum_j$ printed in Luo–Fu Eq. (19), rather than the “all other particles” neighborhood convention of Ref. [4]. The literal physical features v_i^0, e_{ij}^0 are reinserted after every iteration.

11 Backflow, orbital transformation, and determinant

The complex backflow matrix is implemented as a real, bias-free $32 \rightarrow 4$ linear map. Its first and last two outputs are the real and imaginary parts of a two-component displacement,

$$\delta \mathbf{r}_i(X) = W V_i^2, \quad \tilde{\mathbf{r}}_i = \mathbf{r}_i + \delta \mathbf{r}_i. \quad (41)$$

There is no tanh, clipping, or external backflow scale. The fixed mean-field coefficients are evaluated only at this transformed coordinate,

$$\varphi_{\mathbf{k}i}(X) = \sum_{\mathbf{G}} u_{1\mathbf{k}\mathbf{G}}^l \exp[i(\mathbf{k} + \mathbf{G}) \cdot (\mathbf{r}_i + \delta \mathbf{r}_i(X))]. \quad (42)$$

Consequently, backflow does not update $u_{1\mathbf{k}\mathbf{G}}^l$. The factors $\exp[i(\mathbf{k} + \mathbf{G}) \cdot \delta \mathbf{r}_i]$ are generated only by changing the coordinate argument.

The neural orbital transformation uses the Cartesian two-component mesh momentum and the final node feature,

$$[V_i^2, k_x, k_y] \in \mathbb{R}^{34} \xrightarrow{34 \rightarrow 32} \text{GELU} \xrightarrow{32 \rightarrow 2} (J_{\mathbf{k}i}^R, J_{\mathbf{k}i}^I), \quad (43)$$

$$J_{\mathbf{k}i} = J_{\mathbf{k}i}^R + i J_{\mathbf{k}i}^I, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i} \varphi_{\mathbf{k}i}. \quad (44)$$

The direct complex output is not converted to $1 + \alpha \Delta J$. Equation (44) is implemented once: J is not included inside φ and then multiplied again. A unit test that sets $J_{\mathbf{k}i} = 1/2$ verifies $\det Q = 2^{-N_e} \det \varphi$ to numerical precision.

For the Luo–Fu full- M state,

$$\Phi_{\gamma i}(X) = \sum_{\mathbf{k}} M_{\gamma \mathbf{k}} Q_{\mathbf{k}i}(X), \quad (45)$$

$$\Psi_M(X) = \det_{\gamma i} \Phi_{\gamma i}(X), \quad M \in \mathbb{C}^{N_e \times 9}. \quad (46)$$

All graph, W , J , and complex M parameters are trainable. Only one final generalized Slater determinant is used. Because M is trainable and dense, the Cauchy–Binet expansion of $\det(MQ)$ generally contains several total-momentum sectors.

Version 4 starts from the same single dense M but restores exact total momentum by averaging the complete determinant over the nine translations T of the 3×3 torus. Let $D_T = \text{diag}[\exp(i\mathbf{k} \cdot \mathbf{T})]$ act on the momentum rows of Q . The new raw state is

$$\Psi_{\Gamma}^{\text{v4}}(X) = \frac{1}{9} \sum_T \det[MD_T Q(X)]. \quad (47)$$

Only one $N_e \times 9$ matrix M is stored and optimized. The graph and Q are evaluated once for X ; the nine inexpensive contractions and $N_e \times N_e$ determinants implement the group average. There is no list of momentum combinations and no Monte Carlo sampling over such combinations.

Let $\mathcal{C}_{\Gamma}^{(N_e)}$ contain the N_e -momentum subsets with total momentum Γ . Cauchy–Binet makes the many-body content explicit:

$$\Psi_{\Gamma}^{\text{v4}}(X) = \sum_{I \in \mathcal{C}_{\Gamma}^{(N_e)}} \det(M_{:,I}) \det(Q_{I,:}(X)). \quad (48)$$

Indeed, $9^{-1} \sum_T \exp(i\mathbf{K}_I \cdot \mathbf{T})$ is one for $\mathbf{K}_I = \Gamma$ and zero otherwise. Equation (47) is therefore an exact Γ -sector superposition rather than a one-hot determinant, while its coefficients are the structured minors of a single-particle matrix. For $N_e = 3$ and 6, the implementation was verified numerically against the explicit complete Γ -minor sum.

12 Complete-wavefunction outer C_3 projection

The outer construction applies no internal Reynolds average. In v4 its raw state is $D_{\theta}^{\text{v4}}(X) = \Psi_{\Gamma}^{\text{v4}}(X)$ from

Eq. (47), with an unrestricted Luo–Fu message-passing backflow and J network. Thus neither an individual displacement nor an individual generalized orbital is forced to be a C_3 eigenobject, while the complete raw state remains in the translation sector $\mathbf{K}_{\text{tot}} = \Gamma$.

The physical continuum action includes the layer-dependent Bloch-gauge sewing. With the code’s active-rotation convention, define

$$(\mathcal{C}^a D_\theta^{v4})(X) = G_a(X) D_\theta^{v4}(C_3^a X), \quad (49)$$

where $G_0 = 1$ and G_a is the accumulated product of the physical layer-gauge factor at each preceding point on the configuration orbit. For $\omega = \exp(2\pi i/3)$, the complete-state projector is

$$\begin{aligned} \Psi_{\Gamma, m, \theta}^{v4}(X) &= P_m D_\theta^{v4}(X) \\ &= \frac{1}{3} \sum_{a=0}^2 \omega^{-ma} G_a(X) \\ &\quad \times D_\theta^{v4}(C_3^a X), \quad m = 0, 1, 2. \end{aligned} \quad (50)$$

The three independently optimized sectors have exactly the same trainable parameter counts as full M ; projection adds no trainable parameter. Every raw branch is a complete graph/determinant evaluation with shared parameters.

Crucially, all stochastic quantities use Eq. (50), not one raw branch. Metropolis ratios are $|\Psi_m(X')/\Psi_m(X)|^2$, every kinetic or tunnelling local-energy term contains the complete ratio $\Psi_m(X')/\Psi_m(X)$, and SR uses $\mathcal{O}_\mu(X) = \partial_{\theta_\mu} \log \Psi_m(X)$. Setting `outer_c3_projector=False` recovers the v4 Γ -projected wavefunction exactly.

Before production, targeted random-configuration tests confirmed nonzero $m = 0, 1, 2$ sectors, the character relation $\Psi_m(C_3 X)/\Psi_m(X) = \omega^m$, and the projector algebra $P_m P_n = \delta_{mn} P_m$ to double-precision tolerance. Disabling the outer projector recovers the raw v4 state exactly. These tests isolate the outer-projection logic from the

13 Initialization and trainable parameter count

Every affine neural-network kernel uses JAX LeCunNormal, `std = 1/√fan in`, with zero biases. The auxiliary h^0 convention was specified above. For full- M , the real and imaginary components are independently initialized with half-scale LeCunNormal,

$$\mathbb{E}[(\Re M_{\gamma k})^2] = \mathbb{E}[(\Im M_{\gamma k})^2] = \frac{1}{2N_k}, \quad \mathbb{E}|M_{\gamma k}|^2 = \frac{1}{N_k}. \quad (51)$$

Thus every momentum minor is accessible at step zero and there is no preferred Γ subset. Ref. [1] states global LeCunNormal network initialization but does not separately specify M ; the complex extension and variance split above are therefore the present user-requested convention, not an attributed statement from the paper.

The mean-field orbital source is always band 1. However, because W and J themselves are LeCunNormal rather than forced to $W = 0, J = 1$, the step-zero many-body state is not claimed to equal the undressed bare Slater determinant pointwise. This distinction is recorded to make the initialization auditable.

Table 4: Trainable real parameter counts. Version 4 uses the same single complex $N_e \times 9$ matrix M as full M ; finite-group projections add no parameters.

filling	N_e	full M	v4 $P_\Gamma[M]$	v4 $P_m P_\Gamma[M]$
1/3	3	78,154	78,154	78,154
2/3	6	78,208	78,208	78,208

The translation and external P_m projectors share the same raw parameters among all translated or rotated branches and add no trainable parameter.

14 Hamiltonian and local-energy estimator

The VMC objective uses the complete continuum Hamiltonian at $\theta = 3^\circ$, screening parameter $\epsilon = 5$, and a 3×3 torus. It is not the one-band or five-band projected ED Hamiltonian. The local energy contains the kinetic term, layer-dependent moiré potential, spatially varying inter-layer tunnelling, and periodic Coulomb interaction,

$$E_{\text{loc}}(X) = \frac{(H\Psi)(X)}{\Psi(X)} = E_{\text{kin}} + E_{\text{moiré}} + E_{\text{tun}} + E_C. \quad (52)$$

The Coulomb sum uses reciprocal-shell cutoff 8, dielectric $\epsilon = 5$, and removes the $\mathbf{q} = 0$ neutral-background component. Tunnelling is evaluated by flipping one layer index and multiplying the exact wavefunction ratio.

Cartesian derivatives of the complete complex state use the three-point central stencil

$$\partial_a \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) - \Psi(X - h\hat{a})}{2h\Psi(X)}, \quad (53)$$

$$\partial_a^2 \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) + \Psi(X - h\hat{a}) - 2\Psi(X)}{h^2\Psi(X)}. \quad (54)$$

Here $h = 1.5 \times 10^{-3} a_M$. Periodic wrapping is applied after every coordinate displacement. A previous $h, h/2, h/4$ audit is not reused as a validation of the new checkpoints; it must be repeated after convergence.

14.1 Raw-coordinate and full-wavefunction Hamiltonian audit

The production JAX path is `src/jax_neural_bloch.py:184-248`; the independent PyTorch reference is `src/continuum_vmc.py:96-206`.

Both implementations first form $\mathbf{r}_i$ from the input *raw positions*. Consequently,

$$V_M(X) = -2V \sum_{i,a} \cos(\mathbf{g}_a \cdot \mathbf{r}_i + \phi_{l_i}), \quad (55)$$

$$V_C(X) = \sum_{i < j} \sum_{\mathbf{q} \neq 0} v_{\mathbf{q}} \cos[\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)] \quad (56)$$

never see $\delta \mathbf{r}_i$. For tunnelling, $t(\mathbf{r}_i)$ is evaluated at that same raw coordinate, while its numerator is the complete $\Psi(X_{l_i \rightarrow 1-l_i})$ obtained by a new `logpsi` call. The kinetic ratios similarly use new complete calls,

$$R_{i\mu}^{\pm} = \frac{\Psi(X_{\mathbf{r}_i \rightarrow \mathbf{r}_i \pm h\hat{\mu}})}{\Psi(X)}, \quad (57)$$

so each displaced call rebuilds the graph and recomputes every particle's $\delta \mathbf{r}_j$, J_{kj} , the MQ contraction, and the determinant. No backflow displacement is cached by the Hamiltonian.

As a final energy-zero and convention check, a fixed-state regression sets $\delta \mathbf{r} = 0$, $J = 1$, and a fixed M selecting a known two-orbital bare Bloch Slater determinant. The four local-energy components are then compared with an independent finite-basis evaluation. This test uses 4,096 walkers, 120 burn sweeps and 10 measurements and is never used for an optimization update.

Table 5: Fixed-state Hamiltonian regression in meV for the complete many-body state. Parentheses are Monte Carlo standard errors.

term	finite-basis	Monte Carlo	difference
kinetic	39.4207	39.3467(3343)	-0.0740
moiré	-38.5168	-38.7739(5347)	-0.2571
tunnelling	-29.9435	-29.6118(2984)	+0.3317
Coulomb	-9.6769	-10.1671(3354)	-0.4902
total	-38.7164	-39.2060(3354)	-0.4896

Every component and the total agree within 1.5 quoted standard errors. The regression therefore finds no common Hamiltonian normalization, Coulomb background, tunnelling, or energy-zero mismatch at its present precision.

A targeted regression test in `tests/test_jax_neural_bloch.py` instruments the real `_encoded_graph` function. One local-energy call produces four complete graph evaluations: base X with shape $(B, N_e, 2)$, all plus and all minus configurations with shape $(2N_e B, N_e, 2)$ each, and all one-particle layer flips with shape $(N_e B, N_e, 2)$. It checks every displaced coordinate and every flipped layer against an independently constructed array. The test passes, as does the JAX-PyTorch full-local-energy parity test to 2×10^{-7} meV absolute tolerance. No Hamiltonian input-routing error was found in this audit.

15 MCMC and natural-gradient optimization

There are 4,128 simultaneous walkers. Initial fractional coordinates are uniform on the torus and layer indices are uniform in $\{0, 1\}$. Each sweep proposes, for every walker, one random-particle Gaussian displacement with standard deviation 0.06 in fractional supercell coordinates, followed by one random-particle layer flip. Both use the Metropolis ratio $\min[1, |\Psi(X')/\Psi(X)|^2]$. The present jobs use 300 burn-in sweeps and two sweeps per update. Acceptance for coordinate and layer moves is recorded separately.

For real parameters θ_a ,

$$O_a(X) = \partial_{\theta_a} \log \Psi_{\theta}(X), \quad (58)$$

$$S_{ab} = \text{Re} [\langle O_a^* O_b \rangle - \langle O_a^* \rangle \langle O_b \rangle], \quad (59)$$

$$F_a = 2 \text{Re} \langle (E_{\text{loc}} - \langle E_{\text{loc}} \rangle) O_a^* \rangle. \quad (60)$$

The update is

$$(S + \lambda I) \Delta \theta = F, \quad \theta \leftarrow \theta - \eta \Delta \theta, \quad (61)$$

with ordinary, unpreconditioned CG. JAX JVP/VJP operations apply the metric matrix-free. The retained runs check at every Krylov step from iteration 20. For the v4 outer runs at both fillings, checks occur at iterations 20, 25, 30, ..., and additionally whenever the recursive residual predicts convergence or the iteration cap is reached. Acceptance always uses the explicitly recomputed authoritative residual

$$r_{\text{rel}}^{\text{true}} = \frac{\|F - (S + \lambda I) \Delta \theta\|_2}{\|F\|_2} \quad (62)$$

with an additional Fisher-vector product. At a check, the solve stops when $r_{\text{rel}}^{\text{true}} < 0.02$ or at 60 CG iterations. At the cap, an update is accepted only if $r_{\text{rel}}^{\text{true}} < 0.05$ and $F^T \Delta \theta > 0$. Otherwise the parameter update is rejected and the following step uses the recovery values $\lambda = 0.03$ and $\eta = 10^{-3}$. After two consecutive stable recovery solves, the protocol returns to $\lambda = 0.01$ and $\eta = 2 \times 10^{-3}$. Every trace stores the true residual, $F^T \Delta \theta$, force and direction norms, acceptances, and whether the update was applied. This is an adaptive truncated natural gradient, not a claim that the 10^{-3} linear-system tolerance has been reached.

Table 6: Optimization and numerical settings. Quantities not specified by Ref. [1] are explicitly marked as local choices.

quantity	current value	provenance
width / message iterations	32 / 2, unshared	Ref. [1], Appendix B
orbital MLP hidden width	32, GELU	Ref. [1], Appendix B
final determinants	1	local controlled choice
optimizer / implementation	natural gradient / JAX 0.11.1	Ref. [1]
learning rate	2×10^{-3}	Ref. [1]
walkers per update	4,128	Ref. [1]
affine-kernel initialization	LeCunNormal	Ref. [1]
h^0 initialization	random, trainable	random: Ref. [1]; law/sharing: local
full- M initialization	complex LeCunNormal	user choice; paper unspecified
node aggregation	$\sum_j m_{ij}$, including self	Ref. [1], Eq. (19)
updates in present experiment	120	user-specified finite run
paper long-run statement	1,000 for clusters $> 3 \times 3$	Ref. [1]
burn / sweeps per update	300 / 2	local choice
proposal scale	0.06 fractional	local choice
SR damping	10^{-2}	local choice
CG range / true-residual rules	20–60 / 0.02 stop, 0.05 accept	user-specified
wavefunction / SR batch	258 / 258	local memory choice
local-energy batch	32	local memory choice
finite-difference step	$1.5 \times 10^{-3} a_M$	local choice
random seed	26090413 (1/3); 26090423 (2/3)	local choice
precision	JAX 64 bit	local numerical choice

All 4,128 configurations are used in the energy, force, and metric estimate at each update, with sector-specific importance weights during the v4 shared-pool phase. The printed within-step SEM of private chains treats walker values as an instantaneous ensemble diagnostic; it is not an independent uncertainty. Checkpoints at initialization, every 10 updates, the latest completed update, and update 120 store both flattened parameters, optimizer recovery state, sampler RNG, and walker coordinates/layers. Interrupted jobs resume this complete state; no validation configuration ever enters an update.

16 Implementation checks completed before production

Previously completed architecture, parity, observable, and local-energy checks verify:

1. particle exchange changes the determinant sign and no other factor;
2. J_{ki} is applied exactly once;
3. fixed Bloch coefficients are absent from trainable parameters;
4. shifted-coordinate Bloch evaluation agrees with an explicit plane-wave calculation;
5. PyTorch and JAX $\log \Psi$ agree at identical parameters;
6. complete JAX and PyTorch local energies agree;
7. v4 is invariant under both primitive translations, uses exactly one trainable dense M , and agrees with the explicit complete Γ -minor sum to numerical precision;
8. the complete-state C_3 projector satisfies character, idempotency, orthogonality, nonzero-sector-norm and projector-disabled v4 regression tests before its three-sector production run;

9. matrix-free force and Fisher-vector products agree with an explicit score-Jacobian calculation;
10. checkpoint export between JAX and PyTorch is lossless;
11. full- M is reproducibly random, complex, and not one-hot;
12. h_i^0, h_{ij}^0 are nonzero trainable parameters; and
13. a one-particle graph receives its nonzero $j = i$ message in Eq. (19);
14. the 1-RDM uses full replaced-configuration ratios and a fixed bare projector, and its bare-Slater full-basis trace is N_e ; and
15. kinetic displacements and tunnelling flips rebuild the complete JAX graph, while diagonal Hamiltonian factors use raw coordinates.

A one-update end-to-end smoke test also passed MCMC, local-energy evaluation, truncated SR, residual recording, parameter update, and checkpoint writing. These checks validate implementation consistency, not physical convergence. For v4, the focused numerical tests also passed, including parameter-count identity with full M , JAX–PyTorch parity for raw and outer states, the finite-translation character, and the Cauchy–Binet identity. The shared-pool tests include weighted complex SR/Jacobian parity, the uniform-weight limit, amplitude-invariant mixture normalization, disconnected-overlap rejection, and sparse true-residual checks. A shared-pool–fork–private-chain smoke test verified identical initial parameters, separate accepted updates, identical fork pools and distinct branch RNGs.

17 Controlled v4 single- M projection experiments

The historical full- M jobs provide the unrestricted baseline. The two v4 $P_\Gamma[M]$ jobs optimize the exactly translation-projected single- M state for 120 updates with 4,128 persistent walkers. The v4 outer runs at both $\nu = 1/3$ and $2/3$ use three same-seed parameter arrays, with seed 26090413 and 26090423 respectively. The arrays update independently from step one, targeting

$$\mathcal{L} = \frac{1}{3} \sum_{m=0}^2 \frac{\langle \Psi_m(\theta_m) | H | \Psi_m(\theta_m) \rangle}{\langle \Psi_m(\theta_m) | \Psi_m(\theta_m) \rangle}. \quad (63)$$

During the first 20 updates, each sector contributes 1,376 persistent walkers to a common 4,128-configuration pool. Every sector is evaluated on the full pool. MBAR bridge equations estimate relative normalization constants and sector importance weights [2]; energy, force and metric centering all use these weights. This samples a probability mixture, not the coherent superposition density. The

equal factor $1/3$ is canceled between each sector's force and metric, retaining the original per-sector damping convention. The reported weight ESS measures concentration, not time-series decorrelation. Finite-sample normalization and MCMC mixing remain limitations. The exact last update-20 parameter state is retained, without seed screening or best-checkpoint selection. Each branch receives a copy of all 4,128 configurations and its own optimizer state, thermalizes for 300 proposal sweeps with a distinct RNG, and performs 100 private updates. The final-ten training mean determines the displayed sector ordering.

Table 7: Authoritative local calculations used in v4. Both fillings' outer runs use independent parameters and shared mixture sampling for updates 1–20.

ν	ansatz	K_{tot}	result-directory suffix
1/3	full M	mixed	gpu_vmc_adaptive_cg120_seed260904/nu1of3_full_m
1/3	v4 $P_{\Gamma}[M]$	Γ	translation_projected_m_rank1_9cell_seed260904/nu1of3
1/3	v4 $P_0 P_{\Gamma}[M]$	$\Gamma, m = 0, 1, 2$	v4_outer_c3_9cell/nu1of3/p{0,1,2}
2/3	v4 $P_m P_{\Gamma}[M]$	$\Gamma, m = 0, 1, 2$	v4_outer_c3_9cell/nu2of3/p{0,1,2}
2/3	full M	mixed	gpu_vmc_adaptive_cg120_seed260904/nu2of3_full_m
2/3	v4 $P_{\Gamma}[M]$	Γ	translation_projected_m_rank1_9cell_seed260904/nu2of3

17.1 Training-chain diagnostics

Table 8 summarizes the final ten attempted updates for each frozen trace. The mean and RMS describe the persistent training chain and are not independent energy estimates. A rejected update contributes its sampled local energy to the diagnostic trace but leaves the parameters unchanged. Rejection counts for the new v4 runs are reported directly in Table 8.

Table 8: Final-ten training diagnostics. Energies are meV per particle; RMS is training-chain scatter.

ν	ansatz	$\langle E/N_e \rangle$	min. E/N_e	RMS	$\langle r_{\text{rel}} \rangle$	$\langle N_{\text{CG}} \rangle$	reject ₁₂₀
1/3	full M	-36.2595	-36.3078	0.0286	0.0187	53.2	0
1/3	v4 $P_{\Gamma}[M]$	-37.2576	-37.3073	0.0365	0.0175	37.3	0
1/3	v4 $P_0 P_{\Gamma}[M]$	-37.3215	-37.3709	0.0244	0.0169	22.2	0
1/3	v4 $P_1 P_{\Gamma}[M]$	-32.0815	-32.1280	0.0350	0.0184	54.3	0
1/3	v4 $P_2 P_{\Gamma}[M]$	-36.1176	-36.1784	0.0349	0.0166	31.3	0
2/3	v4 $P_0 P_{\Gamma}[M]$	-52.5970	-52.7118	0.0581	0.0174	43.1	0
2/3	v4 $P_1 P_{\Gamma}[M]$	-51.6311	-51.6856	0.0413	0.0176	46.5	0
2/3	v4 $P_2 P_{\Gamma}[M]$	-52.9124	-53.0154	0.0616	0.0173	38.7	0
2/3	full M	-52.1216	-52.1622	0.0376	0.0402	51.7	18
2/3	v4 $P_{\Gamma}[M]$	-52.7486	-52.9033	0.0630	0.0416	53.7	11

The v4 $P_{\Gamma}[M]$ final-ten position/layer acceptances are 0.673/0.622 and 0.649/0.617 at $\nu = 1/3$ and $2/3$. At $\nu = 1/3$, outer P_0, P_1, P_2 give 0.671/0.627, 0.675/0.559 and 0.678/0.530, with mean local-energy variances 54.4, 113.8 and 69.9 meV². Thus a smooth energy trace alone does not establish a zero-variance eigenstate. At $\nu = 2/3$, the outer position/layer acceptances are 0.653/0.607, 0.659/0.616 and 0.652/0.616, with variances 648.3, 682.2 and 643.7 meV². Thus a smooth energy trace does not establish a zero-variance eigenstate.

17.2 Training-tail comparison

Following the optimization convention of Ref. [1], the public energy comparison uses the persistent training ensemble. For each trace we average the final ten measurements and quote their RMS fluctuation; the RMS is a correlated training-chain stability diagnostic, not an independent standard error. At $\nu = 1/3$, P_0 is lowest among the three v4 outer branches, with $\langle E/N_e \rangle_{\text{tail}} = -37.3215$ meV and $\Delta E_{5b} = +0.0718$ meV. The v4 $P_{\Gamma}[M]$ state gives -37.2576 meV. The P_1 branch remains much higher, while P_2 lies 1.2039 meV per particle above P_0 . At $\nu = 2/3$, P_2 is lowest at -52.9124 meV per particle; P_0 and P_1 are higher by 0.3154 and 1.2813 meV. The v4 $P_{\Gamma}[M]$ value is -52.7486 meV, while five-band ED gives -52.7255 meV. These remain correlated optimization-chain statistics rather than independent standard errors. Completion of 120 updates does not establish convergence or seed robustness.

Figure 6 combines the complete nine-cell comparison and all three outer-sector traces on one full page. Its first two rows show optimization, one-five-band ED comparison and bare-band weights; its last two rows show the outer-sector descent, final-ten offsets and timing. CG stability is deliberately omitted from this page and collected for both system sizes in Fig. 11. Rejection statistics remain in Table 8. At $\nu = 1/3$, all displayed nine-cell v4 runs have zero rejected updates. A rejected update leaves the parameters unchanged, although its sampled local energy remains in the diagnostic trace.

Figure 6(i,l) compares the nine-cell QNS runs with five-band ED restricted to total momentum Γ , not the complete Fig. 2–3 production workflow. The measured $\nu = 2/3$ ED job took 660.453 s (11.0 min), with a 905,100-dimensional sector, four eigenpairs, 12 configured OpenMP threads and no GPU. It loads a cached five-band operator and starts from the embedded three-band ground state. The separately recorded operator setup took 17.129 s and is excluded, as are the preceding lower-band solves. The 659.286 s internal timer includes cache loading, the eigensolve, residual and overlap checks; the 660.453 s process timer additionally includes startup and output. Neither is a pure eigensolver-kernel timer. CPU process time was not measured. At $\nu = 1/3$, a dedicated five-band Γ -only timing rerun gives a median process wall time of 0.349 s across five fresh processes (range 0.347–0.389 s); median CPU process time is 1.585 s. The sector dimension is 1,590. It uses the cached interaction, embedded three-band initial vector and the same four-eigenpair settings as $\nu = 2/3$ (tol = 10^{-7} , ncv = 64, maximum 400 iterations). Median internal and eigensolver-only times are 0.153 s and 0.104 s; the process timer additionally includes imports, reference checks and output. All four eigenvalues agree with the saved production results within 10^{-6} meV; existing states are not overwritten. Filesystem caches were not flushed. This supplementary median is not a contemporaneous repeated benchmark of the historical $\nu = 2/3$ job. Berry curvature, other momentum

sectors and the 27-cell spectrum are excluded. The full- M and v4 $P_\Gamma[M]$ times are reported individually, not as a two-filling aggregate: 480/302.833 s at $\nu = 1/3$ and 1,467/1,091.144 s at $\nu = 2/3$. They include startup, initial thermalization and 120 training updates; optional validation, checkpoint export and observables are excluded. Full M mixes total momenta and is an unrestricted baseline, whereas both v4 projected workflows stay exactly in total Γ . The ED calculation returns four eigenpairs, whereas QNS supplies three finite-training candidate sector states in the full continuum ansatz; the different accuracy and basis truncation preclude a matched-accuracy speedup claim. Wall time measures elapsed production time, while accumulated device time sums simultaneous workers and therefore records resource consumption. The ED rows used no GPU; the QNS rows used one A30 and one 16-vCPU affinity per concurrent job. Software installation, smoke tests, figure rendering and $\text{\TeX}$ compilation are excluded. The v4 $\nu = 1/3$ outer workflow took 22.81 minutes and consumed 0.559 allocated GPU hours; its private branches took 5.97, 13.33 and 7.31 minutes for P_0, P_1, P_2 . GPU-hours count lane/worker allocation, not measured kernel execution. Since the workflow produces three symmetry-sector candidates, the displayed amortized cost is the complete wall/device cost divided by three. Shared-stage costs are included before division. This measures workflow throughput under the two-GPU schedule, not the latency of an isolated eigenstate calculation. The candidates have not been certified as converged eigenstates. At $\nu = 2/3$, the v4 workflow took 87.84 minutes and 2.227 allocated GPU hours; private P_0, P_1, P_2 branches took 34.14, 38.33 and 30.57 minutes. The corresponding amortized cost is 29.28 wall-minutes and 0.742 allocated GPU hours per candidate. The complete v4 observable/export queue took 47.54 wall-seconds and 0.026 allocated GPU hours for eight new states. The measured host-process CPU time (user plus system) is 320.61 and 497.42 s for the two single-state v4 $P_\Gamma[M]$ runs, and 1,049.04 and 1,960.08 s for the complete outer workflows at $\nu = 1/3$ and $2/3$, respectively.

Table 9: Nine-cell ED/QNS timing. ED is restricted to five bands and Γ . GPU-hours are allocation sums; CPU seconds are measured process consumption. The $\nu = 1/3$ ED entry is a five-process median; $\nu = 2/3$ is the historical job. Each amortized outer row divides the complete three-sector training cost by three; other workflow rows retain their original scopes.

workflow	scope	production resources	wall time	device time
ED, 5 bands, Γ	$\nu = 1/3$, four eigenpairs	12 OpenMP threads	0.349 s	1.585 CPU s
ED, 5 bands, Γ	$\nu = 2/3$, four eigenpairs	12 OpenMP threads	11.0 min	CPU time not recorded
QNS full M training	$\nu = 1/3$, one state	A30	480 s	0.133 GPU h
QNS v4 $P_\Gamma[M]$ training	$\nu = 1/3$, one state	A30	302.833 s	0.084 GPU h
QNS full M training	$\nu = 2/3$, one state	A30	1,467 s	0.408 GPU h
QNS v4 $P_\Gamma[M]$ training	$\nu = 2/3$, one state	A30	1,091.144 s	0.303 GPU h
QNS v4 Outer- C_3 training	$\nu = 1/3$, three sectors	2 $\times$ A30	22.81 min	0.559 GPU h
Outer amortized cost	$\nu = 1/3$, per candidate	same schedule	7.60 min/sector	0.186 GPU h/sector
QNS v4 Outer- C_3 training	$\nu = 2/3$, three sectors	2 $\times$ A30	87.84 min	2.227 GPU h
Outer amortized cost	$\nu = 2/3$, per candidate	same schedule	29.28 min/sector	0.742 GPU h/sector
v4 observables/export	eight new states	2 $\times$ A30	47.54 s	0.026 GPU h

18 Twenty-seven-cell QNS extension

The same three ansatz families were subsequently trained on the 27-cell $3\sqrt{3} \times 3\sqrt{3}$ torus used by the one-band ED calculation in Fig. 3. This extension changes the continuum angle to $\theta = 2.6^\circ$, uses all 27 mesh momenta and replaces the nine-term translation average in Eq. (47) by the exact 27-element finite translation group. The network width and orbital-MLP width are both 64. Consequently the $N_e = 9$ and 18 states contain 571,374 and 571,860 real trainable parameters, respectively. The sampler still contains 4,128 persistent walkers and uses 300 burn sweeps, two sweeps per update and 120 natural-gradient updates. Full M , $P_\Gamma[M]$, and every outer sector use seed 3184. For the outer calculation, the first 20 updates use the shared-mixture pool and the remaining 100 updates use independently evolved parameters and walkers, exactly as in the nine-cell protocol.

Table 10: Twenty-seven-cell final-ten training diagnostics. Energies and RMS are meV per particle. “Reject” counts attempted parameter updates rejected by the explicit true-residual criterion.

ν	ansatz	$\langle E/N_e \rangle$	RMS	reject ₁₂₀
1/3	full M	-46.3697	0.0343	59
1/3	v4 $P_\Gamma[M]$	-46.6770	0.0346	49
1/3	v4 $P_0 P_\Gamma[M]$	-47.5342	0.0347	24
1/3	v4 $P_1 P_\Gamma[M]$	-46.7285	0.0329	32
1/3	v4 $P_2 P_\Gamma[M]$	-46.6239	0.0309	40
2/3	full M	-59.5784	0.0514	94
2/3	v4 $P_\Gamma[M]$	-59.8350	0.0217	92
2/3	v4 $P_0 P_\Gamma[M]$	-60.2690	0.0161	91
2/3	v4 $P_1 P_\Gamma[M]$	-60.4155	0.0391	75
2/3	v4 $P_2 P_\Gamma[M]$	-60.3833	0.0249	85

Figure 7 repeats the complete Fig. 6 layout for 27 cells. Because a five-band 27-cell ED calculation is not available, its energy-comparison panels use the five QNS method labels on the horizontal axis. The one-band ED energies -47.595463 and -57.455730 meV per particle are horizontal dashed references at $\nu = 1/3$ and $2/3$. At $\nu = 1/3$, outer $m = 0$ is lowest and lies 0.0612 meV per particle above the one-band ED reference. At $\nu = 2/3$, outer $m = 1$ is lowest; it and $m = 2$ differ by only 0.0322 meV per particle, below their combined tail scatter. All continuum QNS values at this filling lie below one-band ED because the neural state contains continuum/remote-band freedom absent from the one-band projected Hamiltonian. This is not a variational violation and the dashed line is a reference, not a matched-Hamiltonian bound.

The measured 27-cell wall times are listed in Table 11. The one-band ED entries are cached Γ -sector solves and exclude the full all-momentum spectrum and Berry calcu-

lation. As in the nine-cell figure, the displayed outer cost is the complete two-GPU workflow wall time divided by three because the workflow returns three candidate symmetry sectors. This amortized throughput should not be confused with the latency or GPU allocation of a single isolated run.

Table 11: Twenty-seven-cell ED/QNS timing. QNS rows are 120-update training jobs. Outer wall time is the observed two-GPU critical path; allocated GPU time sums both lanes.

ν	workflow	wall time	allocated GPU time
1/3	one-band ED, Γ	~ 30 s	—
1/3	full M	2.37 h	2.37 GPU h
1/3	v4 $P_\Gamma[M]$	2.38 h	2.38 GPU h
1/3	outer, complete / amortized	14.33 / 4.78 h	22.84 / 7.61 GPU h
2/3	one-band ED, Γ	~ 29 s	—
2/3	full M	10.89 h	10.89 GPU h
2/3	v4 $P_\Gamma[M]$	11.11 h	11.11 GPU h
2/3	outer, complete / amortized	66.66 / 22.22 h	106.58 / 35.53 GPU h
both	observables, ten states	6.16 min	0.194 GPU h

Figure 11 isolates the true CG residuals from both system sizes. The nine-cell $\nu = 1/3$ runs remain near the 0.02 early-stop line, whereas many 27-cell late updates reach the 60-iteration cap and exceed the 0.05 acceptance threshold. This explains the large rejection counts in Table 10 and shows that the present 27-cell energies are finite-optimization diagnostics rather than converged sector eigenvalues.

19 Post-training one- and two-body observables

Figure 8 compares full- M and v4 $P_\Gamma[M]$ at both fillings on 27 cells. Figures 9 and 10 display all three 27-cell externally projected v4 sectors; the final-ten training minimum is selected from the completed local traces and marked in each figure. The panels are regenerated from the saved final training ensembles, with no fresh validation chains. Each contains 4,128 configurations; energy selection and observable estimation use the same production-chain convention as the optimization protocol of Ref. [1]. Both occupations and the primary structure factor are shown at exactly the 27 physical momentum vectors used in the ED figure. A separately evaluated 37-vector grid completes the boundary under C_6 and is retained only to audit rotations of physical continuum Fourier modes. Figure generation checks the 27 main vectors point by point. No observable is C_3 averaged and no ED momentum profile is used in postprocessing.

19.1 Bloch-projected momentum occupations and bare-band weights

To compare like with like, momentum occupation is defined in the same fixed bare-band basis as the one-band ED calculation. For $a = (n, \mathbf{k})$ and $b = (m, \mathbf{q})$, its estima-

tor is

$$\begin{aligned} \rho_{ab} &= \langle c_b^\dagger c_a \rangle \\ &= 2N_e \mathbb{E}_{X \sim |\Psi|^2, i, x'} \left[(\phi_a^0(x_i))^* \phi_b^0(x') \left(\frac{\Psi(X_{i \rightarrow x'})}{\Psi(X)} \right)^* \right]. \end{aligned} \quad (64)$$

The factor two accounts for uniform auxiliary sampling of the two layers. In every ratio, replacing one particle triggers a complete new ansatz call: the code rebuilds the graph and recomputes all particles' $\delta \mathbf{r}_j$, $J_{\mathbf{k}j}$, and either the MQ determinant or all nine translated MD_TQ determinants. The projector is instead the fixed bare function

$$\phi_{n\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{n\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{r}}; \quad (65)$$

no backflow phase is multiplied into it. Thus neural dressing appears once, only through the complete wavefunction ratio.

The raw first-band occupation and the five-band total displayed in Figs. 8–10 are

$$\tilde{n}_1(\mathbf{k}) = \rho_{1\mathbf{k}, 1\mathbf{k}}, \quad \sum_{\mathbf{k}} \tilde{n}_1(\mathbf{k}) = N_e W_1, \quad (66)$$

$$n_{\text{tot}}(\mathbf{k}) = \sum_{n=1}^5 \rho_{n\mathbf{k}, n\mathbf{k}}. \quad (67)$$

For direct shape comparison with one-band ED, the plotted first-band panel is the explicitly conditional normalization

$$n_1(\mathbf{k}) = N_e \frac{\tilde{n}_1(\mathbf{k})}{\sum_{\mathbf{q}} \tilde{n}_1(\mathbf{q})}, \quad \sum_{\mathbf{k}} n_1(\mathbf{k}) = N_e. \quad (68)$$

This uses only the QNS first-band trace and the physical filling; it does not fit the ED values. The raw $\tilde{n}_1$ is saved alongside it. The total panel is not rescaled and therefore also exposes the five-band projected-trace error. Five bands are retained because bands four and five are already at the Monte Carlo estimator's percent-level noise floor; adding still more remote bands would primarily accumulate projection noise in the present sample.

For the selected $\nu = 1/3$, $m = 0$ branch, the filling-normalized band-1 QNS profile has RMS difference 0.073 and Pearson correlation 0.828 with the one-band ED result. The corresponding selected $\nu = 2/3$, $m = 1$ values are 0.079 and 0.742. These are blind comparisons between independently computed profiles, with no affine or pointwise fit. The raw band-1 sums are 8.375 and 16.224, respectively, before the conditional normalization to 9 and 18. No rotation average is applied. On these two selected outer- C_3 states, the largest differences between normalized C_3 partners are 0.055 and 0.163; when the block errors of both partners are combined, the largest deviations are 1.39 and 1.95 standard errors. Thus their visible residual asymmetry is consistent with the finite auxiliary-ratio estimate at the present precision, rather than being removed by symmetry postprocessing.

For the nine-cell band-weight validation, all 4,128 final configurations and 16 auxiliary points per configuration were used to estimate the complete 45×45 five-band-projected matrix. For the 27-cell figures, the same complete-ratio construction evaluates the 135 diagonal elements directly; their momentum-resolved sum gives $n_{\text{tot}}(\mathbf{k})$ without materializing a noisy 135×135 matrix. The band populations are

$$W_n = \frac{1}{N_e} \sum_{\mathbf{k}} \rho_{n\mathbf{k},n\mathbf{k}}. \quad (69)$$

Table 12: Nine-cell five-band-projected 1-RDM validation from final training samples. Band weights are percentages; parentheses are eight-block estimates. ED weights use the exact five-band one-body density matrix.

ν	state	trace	W_1	W_2	W_3	W_4	W_5	min eig.
1/3	five-band ED	3	96.47	2.95	0.52	0.05	0.01	—
1/3	v4 $P_T[M]$	3.089(90)	96.91(1.06)	2.52(89)	-0.28(1.11)	2.62(1.02)	1.18(65)	-0.091
1/3	v4 $P_0P_T[M]$	2.995(116)	95.46(1.54)	3.98(1.19)	-0.03(72)	-0.47(63)	0.90(1.02)	-0.123
1/3	v4 $P_1P_T[M]$	3.125(63)	80.80(77)	16.69(1.18)	3.49(1.12)	1.05(28)	2.14(64)	-0.114
1/3	v4 $P_2P_T[M]$	2.880(68)	95.08(1.06)	2.65(1.30)	-1.51(76)	1.34(90)	-1.57(43)	-0.107
2/3	five-band ED	6	88.95	7.59	2.53	0.76	0.16	—
2/3	v4 $P_T[M]$	6.083(148)	90.07(1.02)	5.85(36)	3.48(1.40)	0.61(71)	1.38(1.22)	-0.150
2/3	v4 $P_0P_T[M]$	6.069(185)	88.76(2.14)	7.55(1.08)	1.65(1.02)	1.62(97)	1.58(74)	-0.173
2/3	v4 $P_1P_T[M]$	6.021(193)	87.15(1.84)	7.83(1.01)	3.58(1.08)	0.09(1.27)	1.70(1.04)	-0.376
2/3	v4 $P_2P_T[M]$	6.100(192)	89.83(1.47)	7.15(90)	2.80(1.38)	0.42(1.02)	1.47(70)	-0.205

Table 13: Twenty-seven-cell diagonal five-band projection from the saved final training ensembles. T_5 is the trace within the five-band subspace and W_n are percentages of the physical particle number; parentheses are eight-block SEM estimates. Negative high-band entries quantify ratio-estimator noise and are not physical occupations.

ν	state	T_5	W_1	W_2	W_3	W_4	W_5
1/3	full M	9.195	88.68(1.77)	10.25(1.87)	4.72(2.44)	-0.79(2.04)	-0.69(1.54)
1/3	$P_T[M]$	8.703	90.89(2.04)	7.50(1.27)	0.63(1.96)	-1.60(2.15)	-0.71(1.94)
1/3	$P_0P_T[M]$	9.098	93.06(1.61)	9.14(1.36)	0.98(1.52)	-0.31(1.10)	-1.78(1.21)
1/3	$P_1P_T[M]$	8.833	88.71(2.00)	10.66(2.26)	1.59(1.17)	-2.54(1.32)	-0.28(2.39)
1/3	$P_2P_T[M]$	7.637	88.16(1.59)	5.48(2.68)	-2.49(2.33)	-4.19(2.09)	-2.10(1.68)
2/3	full M	18.469	88.35(2.93)	5.82(1.98)	5.42(3.40)	0.81(1.47)	2.20(1.39)
2/3	$P_T[M]$	17.399	90.54(2.69)	6.46(2.02)	0.80(2.07)	-1.62(2.60)	0.47(1.06)
2/3	$P_0P_T[M]$	17.154	85.73(1.48)	4.43(1.72)	5.53(2.22)	-0.14(1.37)	-0.25(1.60)
2/3	$P_1P_T[M]$	18.316	90.13(1.89)	10.19(2.66)	3.86(2.14)	-1.56(1.20)	-0.87(1.12)
2/3	$P_2P_T[M]$	17.618	87.84(1.67)	7.13(2.00)	4.49(1.71)	0.35(1.30)	-1.93(1.39)

The nine-cell matrices are Hermitized but are not projected onto the positive-semidefinite cone. Their finite-sampling negative eigenvalues must therefore be inspected before any natural-orbital interpretation. The diagonal band populations are the more robust comparison with the exact five-band ED distributions (96.47, 2.95, 0.52, 0.05, 0.01)% at $\nu = 1/3$ and (88.95, 7.59, 2.53, 0.76, 0.16)% at $\nu = 2/3$. At $\nu = 1/3$, the selected $P_0P_T[M]$ branch has (95.46, 3.98)% in the first two bare bands, while $P_1P_T[M]$ is qualitatively different at (80.80, 16.69)%. At $\nu = 2/3$, the selected $P_2P_T[M]$ branch gives (89.83, 7.15, 2.80, 0.42, 1.47)%, close in its first three bands to the ED distribution. The traces fluctuate around N_e , but the negative high-band estimates and eigenvalues quantify the remaining ratio-estimator noise; they must not be read as physical negative occupations.

The estimator caches $\Psi_m(X)$ once per configuration block and batches the 16 uniform replacements used for

both momentum occupations and band weights. Each replacement still rebuilds the complete graph, all $\delta\mathbf{r}_j$, J , the finite-translation determinant sum, the projected three-branch sum and the wavefunction ratio. On the A30 GPUs, the 4,128-walker measurements are timed independently from training on the scheduled dual-GPU critical path. Training and postprocessing are timed separately in Table 9. The focused bare-Slater regression sets $\delta\mathbf{r} = 0$, $J = 1$, fixes a complex 2×9 matrix M with orthonormal rows, and samples the resulting single determinant. In a complete nine-orbital test basis the exact matrix is $\rho = M^\dagger M^*$, with trace two. With 4,096 walkers, 100 burn sweeps and 32 auxiliary draws, the measured trace is 2.01047 ± 0.01459 , the Frobenius error is 0.05272, the Hermiticity residual is zero, and the minimum sampled eigenvalue is -0.02072. This unit test verifies occupation, Hermiticity, PSD within sampling noise and full-basis trace normalization.

For the 27-cell states, the selected $\nu = 1/3$ outer branch $m = 0$ has estimated first-two-band weights (93.06(1.61), 9.14(1.36))%. The selected $\nu = 2/3$ branch $m = 1$ has $(W_1, W_2, W_3) = (90.13(1.89), 10.19(2.66), 3.86(2.14))\%$. The five-band projected traces range from 7.637 to 9.195 at $N_e = 9$ and from 17.154 to 18.469 at $N_e = 18$. In particular, the $m = 2$, $\nu = 1/3$ trace deficit and several negative remote-band diagonal estimates are larger than in the nine-cell audit. Therefore Fig. 7 displays the raw estimates and their block errors without renormalizing them to 100%; the high-band bars should be interpreted only at the quoted Monte Carlo precision.

19.2 Full structure factor, connected part, and density order

Following the full-structure-factor definition of Luo, Zaklaman and Fu [1], the code first forms the empirical real-space density-pair measure and then Fourier transforms it, without a Bloch-projected operator and without reconstructing it from the 1-RDM:

$$\chi(\mathbf{r}, \mathbf{r}') = \langle \rho(\mathbf{r}) \rho(\mathbf{r}') \rangle, \quad (70)$$

$$S_{\text{full}}(\mathbf{q}) = \frac{1}{N_e} \int d\mathbf{r} d\mathbf{r}' e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')} \chi(\mathbf{r}, \mathbf{r}') = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle}{N_e}, \quad (71)$$

$$S_c(\mathbf{q}) = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle - \langle |\rho_{\mathbf{q}}| \rangle^2}{N_e}, \quad (72)$$

$$m_{\mathbf{q}} = \frac{|\langle \rho_{\mathbf{q}} \rangle|}{N_e}. \quad (73)$$

Here $\rho_{\mathbf{q}} = \sum_i e^{i\mathbf{q} \cdot \mathbf{r}_i}$ includes every particle and both layers. Numerically, the unbinned pair displacements are transformed in small configuration batches, so no real-space histogram or FFT interpolation error is introduced. The 27 plotted $\mathbf{q}$ vectors are pointwise identical to the physical Bloch momenta in the ED figure. This representative set is not a closed geometric point set on the hexagonal boundary: one 120° rotation leaves six points without a physical partner. Its rotational completion has

37 distinct vectors, which the code also evaluates directly as an audit. It neither sums over $\mathbf{q} + \mathbf{G}$ nor substitutes a reciprocal-equivalent boundary value, because the continuum density is not periodic under $\mathbf{q} \mapsto \mathbf{q} + \mathbf{G}$. The ED comparison uses the same $1/N_e$ normalization but a one-band-projected density operator, so agreement is expected first in symmetry and momentum dependence rather than exact high- q amplitude. The new v4 values are evaluated independently in every sector, so differences among the rows reflect different saved wavefunctions rather than duplicated plotting data. The largest nonzero- q density amplitudes are also reported to distinguish a fluctuation peak in S_{full} from a nonzero static density expectation. The real-space panels in Figs. 9 and 10 include the intra-moiré-cell Bloch density and should not be identified with a nonzero CDW order parameter by visual contrast alone.

For the nine-cell validation, the largest nonzero-momentum values of S_{full} are 1.4818 for $P_{\Gamma}[M]$ and 1.4852, 1.1325 and 1.3018 for outer $m = 0, 1, 2$, compared with 1.4381 in five-band ED. At $\nu = 2/3$ the corresponding values are 0.7677 and 0.7377, 0.6832 and 0.7013, compared with 0.6197 in ED. The largest nonzero- q density amplitudes are small: (0.01120, 0.00902, 0.01069, 0.00719) for the $\nu = 1/3$ $P_{\Gamma}, m = 0, 1, 2$ states and (0.00475, 0.00921, 0.00590, 0.00351) at $\nu = 2/3$. Thus the displayed structure-factor maxima are predominantly fluctuation signals on these finite training samples, rather than direct evidence for static density order.

For 27 cells at $\nu = 1/3$, the corrected first-BZ nonzero maxima for full M , $P_{\Gamma}[M]$ and outer $m = 0, 1, 2$ are respectively (1.1575, 1.1722, 1.5222, 1.1937, 1.2116), compared with 1.0571 in one-band ED. At $\nu = 2/3$, the corresponding QNS maxima are (0.6976, 0.7236, 0.7492, 0.7446, 0.7254) and remain much larger than the one-band ED maximum 0.1968. Hence the present 120-update 27-cell states do not reproduce the weak one-band ED $\nu = 2/3$ structure-factor amplitude, although the selected-state momentum dependence is close: its Pearson correlation with ED is 0.984 (0.806 at $\nu = 1/3$). Some difference is physically allowed because the QNS uses the untruncated continuum Hilbert space, but the high rejected-update counts and lack of an independent inference chain prevent attributing the whole discrepancy to band mixing. Figures 8–10 use one shared range covering all ED and QNS values, so neither the K peak nor the corrected boundary response is clipped.

The maximum nonzero density amplitudes $m_{\mathbf{q}}$ are 0.0920 and 0.0277 for the momentum-unrestricted full- M states at $\nu = 1/3$ and $2/3$. After exact translation projection they fall below 0.011 for every $P_{\Gamma}[M]$ and outer state. The repeated intra-cell texture in the density column is therefore mainly the moiré-scale one-body density, not evidence of a large static translation-breaking order parameter. Together with the high 27-cell CG rejection counts, the $\nu = 2/3$ structure-factor discrepancy is a clear target for longer, better-conditioned and multi-seed optimization.

The pair estimator obeys $S(\mathbf{q}) = S(-\mathbf{q})$ configuration by configuration. No C_3 average is imposed: rotational differences on the directly evaluated 37-vector audit grid remain visible as a wavefunction and Monte Carlo diagnostic. We do not apply a reciprocal-lattice wrap when auditing boundary points, because that would test $S(\mathbf{q} + \mathbf{G})$ rather than the physical vector obtained by rotating the dot.

20 Conclusion

The implemented baseline follows the printed Luo-Fu Appendix-B structure: fixed mean-field band-1 plane-wave coefficients, periodic node and edge features, trainable random auxiliary features, two unshared message iterations with the literal self-inclusive $\sum_j m_{ij}$, coordinate backflow, direct complex $J_{\mathbf{k}i}$ applied exactly once, and one final generalized Slater construction. Full M mixes momenta at the single-particle level and therefore generally mixes total-momentum sectors. Version 4 keeps that single dense M and applies the exact finite-translation projector to the complete determinant. Cauchy–Binet proves that the result combines all and only the total- Γ minors with coefficients $\det M_{\cdot, I}$, without enumerating momentum combinations in the implementation.

The outer construction leaves that raw network unrestricted and projects the complete $P_{\Gamma}[M]$ state with P_m . Character and projector-algebra tests pass at double-precision tolerance, all three projected norms are nonzero, and local energy, Metropolis ratios and SR derivatives all use the projected state. It adds no trainable parameter. At nine cells the projected and unprojected models contain the same 78,154 and 78,208 real parameters at $\nu = 1/3$ and $2/3$; the width-64 27-cell counterparts contain 571,374 and 571,860.

On the final-ten training tail at $\nu = 1/3$, v4 $P_{\Gamma}[M]$ reaches -37.2576 meV per particle and outer P_0 is the lowest projected branch at -37.3215 meV. At $\nu = 2/3$, the unrotated v4 state reaches -52.7486 meV; P_2 is the lowest outer branch at -52.9124 meV per particle. These are single-initialization training-chain diagnostics, so they do not establish convergence or resolve seed sensitivity.

On 27 cells, full M and $P_{\Gamma}[M]$ reach -46.3697 and -46.6770 meV per particle at $\nu = 1/3$, while outer $m = 0$ is lowest at -47.5342 meV, 0.0612 meV above the one-band ED reference. At $\nu = 2/3$, the respective values are -59.5784 and -59.8350 meV, and outer $m = 1$ is lowest at -60.4155 meV. The $m = 1, 2$ tail means differ by only 0.0322 meV per particle, but extensive residual-cap rejections prevent interpreting that proximity as a converged topological multiplet. The observed per-state QNS training cost is hours rather than the roughly 30-second cached one-band Γ ED solve; this comparison changes qualitatively for multiband ED, whose 27-cell Hilbert space is not available here.

The 27-cell observables show that the projected $\nu = 1/3, m = 0$ state retains zone-corner structure-factor en-

hancement. At $\nu = 2/3$, however, all QNS structure-factor maxima remain near 0.74–0.75, above the one-band ED maximum 0.197, even though static nonzero- q density amplitudes of all translation-projected states stay below 0.011. Thus the current finite run does not yet reproduce the ED liquid correlations. The complete-ratio one-body spatial correlation and direct two-body structure factors remain useful diagnostics, but projected-trace noise, single-seed optimization and SR instability preclude a natural-orbital or topological claim. The next controlled tests are longer better-conditioned multi-seed optimization and sector-selection replication.

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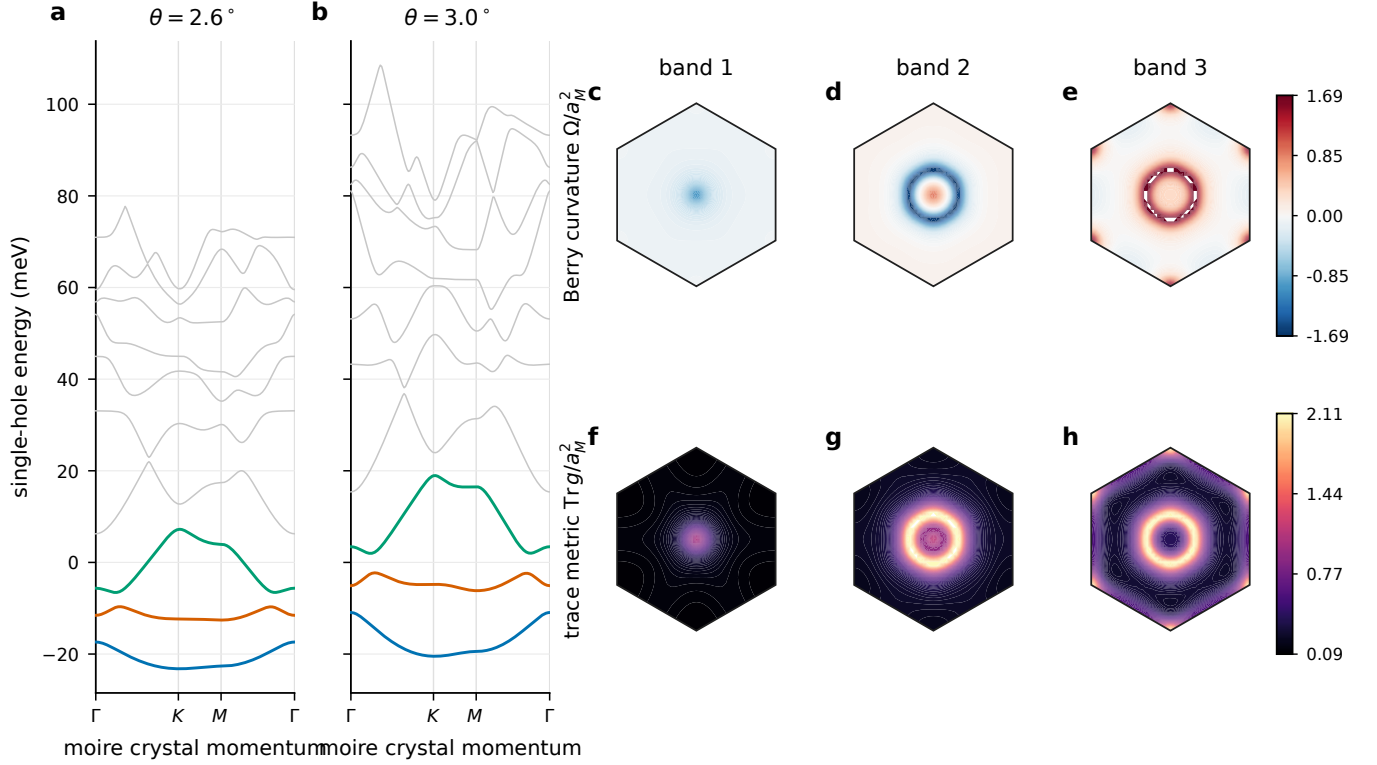


Figure 1: Continuum bands and quantum geometry. (a,b) The first ten moiré bands along Γ - K - M - Γ at $\theta = 2.6^\circ$ and 3.0° ; bands 1–3 are colored blue, orange and green, respectively. (c–e) Berry curvature and (f–h) trace quantum metric for bands 1–3 at $\theta = 2.6^\circ$ across the hexagonal Brillouin zone. Both rows use a common physical color scale with explicit numeric ticks.

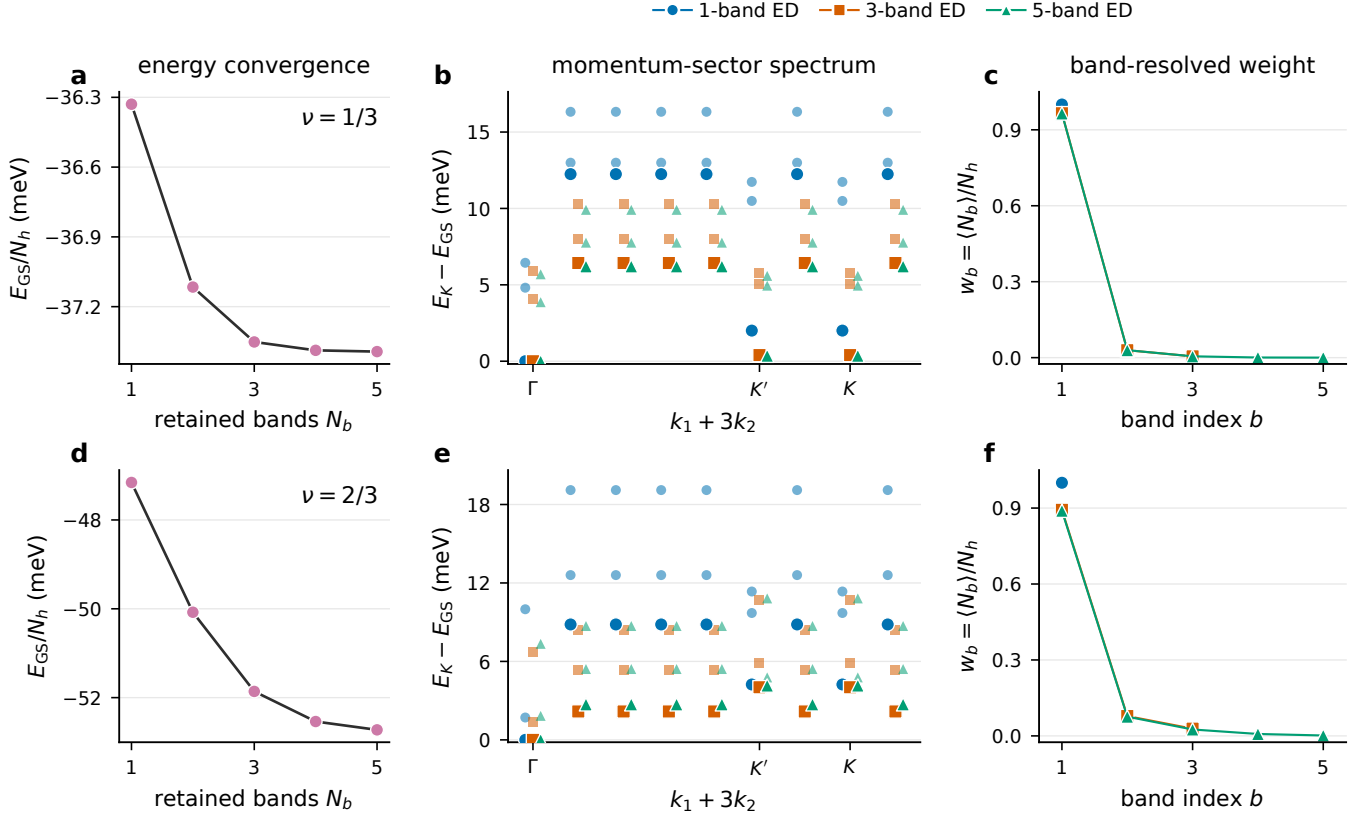


Figure 2: Multiband ED on the 3×3 cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Ground-state energy per hole versus retained bands. (b,e) The lowest three levels in every momentum sector for one-, three- and five-band ED, plotted as $E - E_{GS}$ in meV. Larger opaque markers denote sector ground states; the next two levels are smaller and semitransparent. (c,f) Normalized band-resolved ground-state weights. Charge- $U(1)$, spin- $U(1)$ and translations are exact; C_3/C_6 equivalence accelerates the largest five-band sectors.

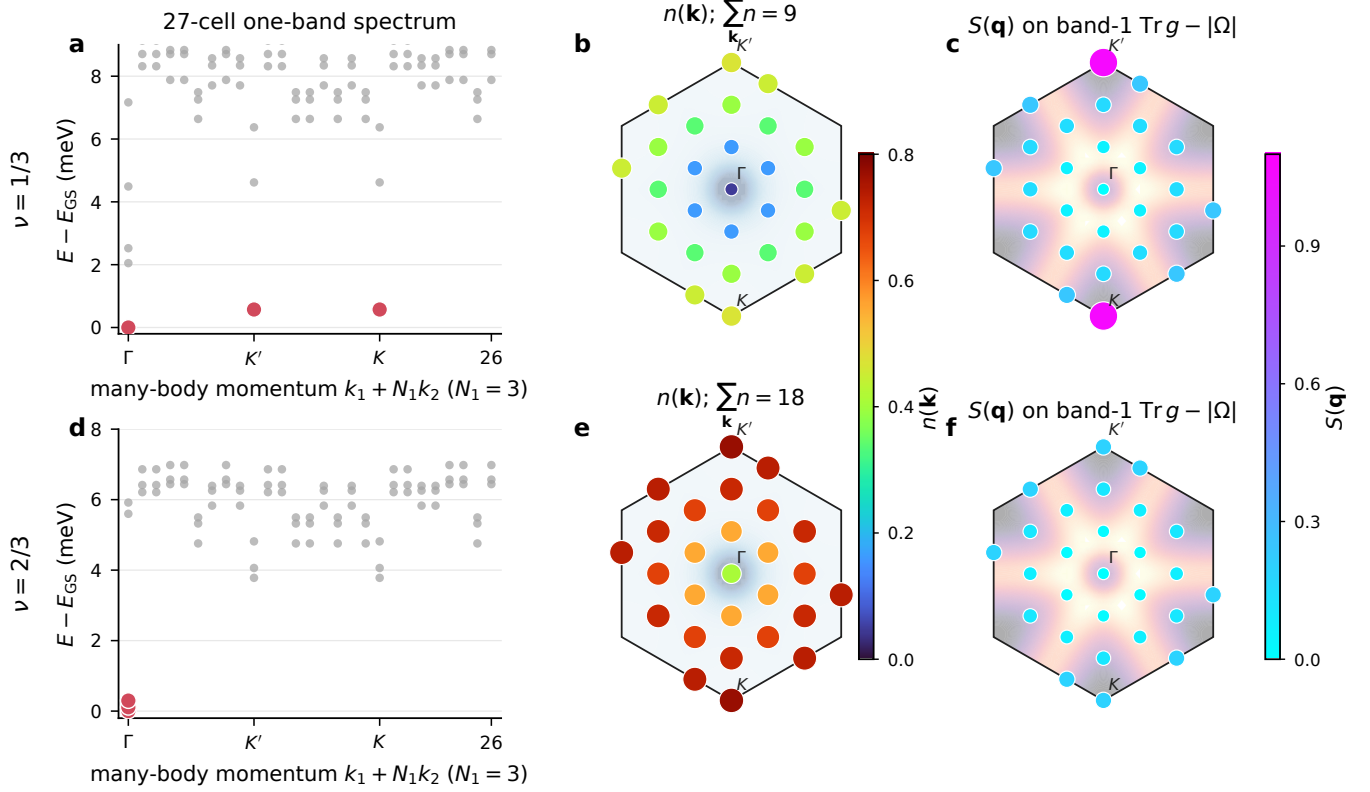


Figure 3: 27-site one-band reproduction of Supplementary Fig. 5. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Momentum-resolved spectra, plotted as $E - E_{GS}$ in meV. Red markers denote the CDW or FCI three-state manifold. (b,e) Ground-manifold averaged $n(\mathbf{k})$ over weak, interpolated band-1 Berry curvature. (c,f) Projected $S(\mathbf{q})$ over the identical weak, interpolated band-1 $\text{Tr } g_1 - |\Omega_1|$ background. Bright marker colorbars encode the observables independently.

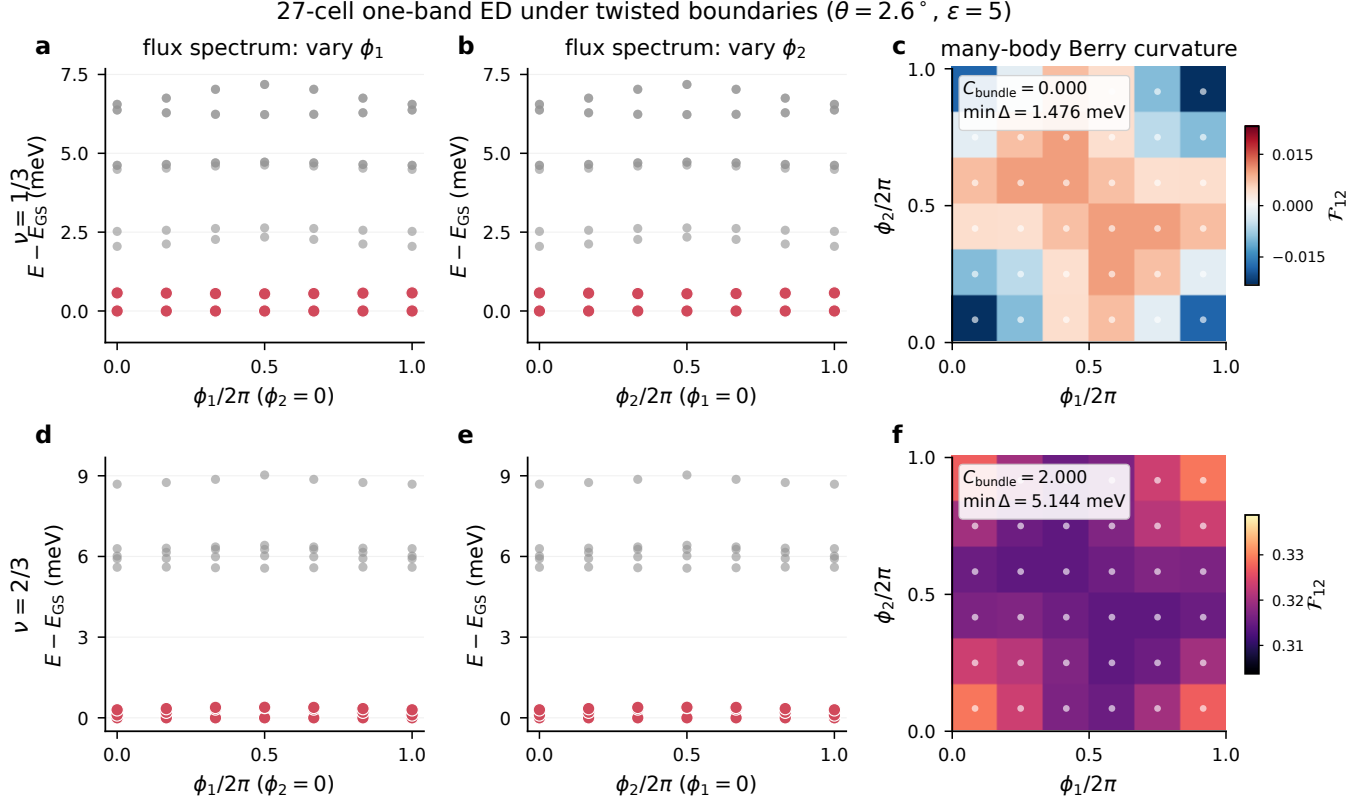


Figure 4: Twisted-boundary ED and many-body topology on the 27-cell one-band cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Spectral flow versus ϕ_1 at $\phi_2 = 0$. (b,e) Spectral flow versus ϕ_2 at $\phi_1 = 0$. The lowest three energies are red. Gray points show additional sampled levels. (c,f) Non-Abelian curvature $\mathcal{F}_{\phi_1\phi_2}$ on the original 6×6 primitive twist grid; white dots mark FHS plaquette centers. The raw plaquette-flux sums give Chern number zero for the CDW and two for the tracked FCI manifold.

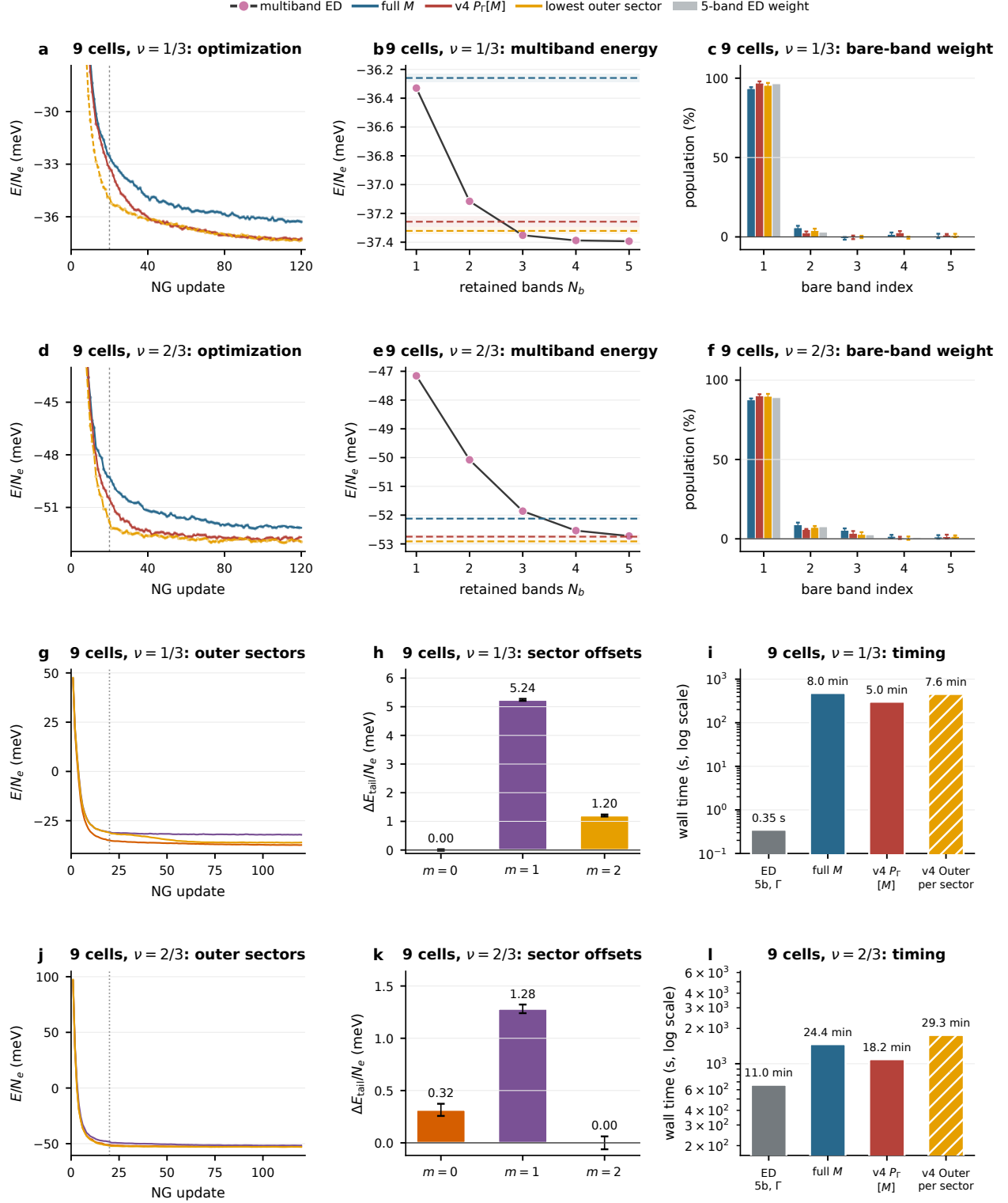


Figure 6: Nine-cell QNS summary. The former optimization and timing figures are combined on one full page. Rows one and two show $\nu = 1/3$ and $2/3$ optimization traces, one-five-band ED energy comparisons and bare-band weights. Rows three and four show all outer- C_3 sector traces, final-ten sector offsets, and the four-method timing comparison. The outer timing is the complete three-sector wall time divided by three. Error bars and shaded energy widths are final-ten RMS fluctuations, not independent uncertainties. The selected outer-state band bars are yellow. CG stability is omitted here and collected in Fig. 11.

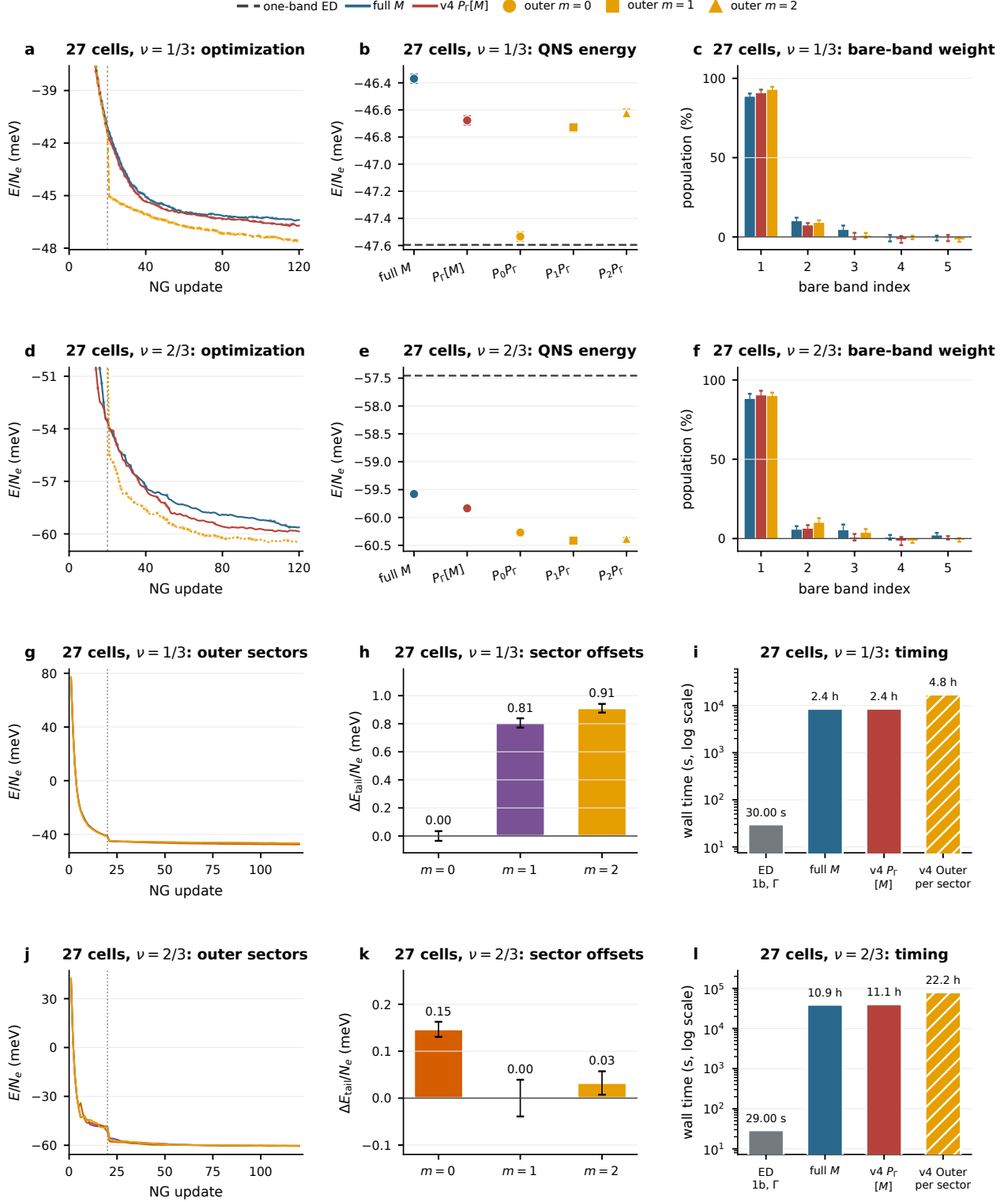


Figure 7: Twenty-seven-cell QNS summary. The layout follows Fig. 6. Since five-band 27-cell ED is unavailable, the energy panels use full M , $P_\Gamma[M]$ and $P_m P_\Gamma[M]$ as categorical abscissa labels; the one-band 27-cell ED energy is a dashed horizontal reference. All three outer-energy markers are yellow, with circle, square and triangle denoting $m = 0, 1, 2$; the selected outer-state band bars use the same yellow. Rows three and four resolve the three outer sectors and report measured training times. The hatched outer cost is the complete two-GPU critical path divided by the three resulting candidate states.

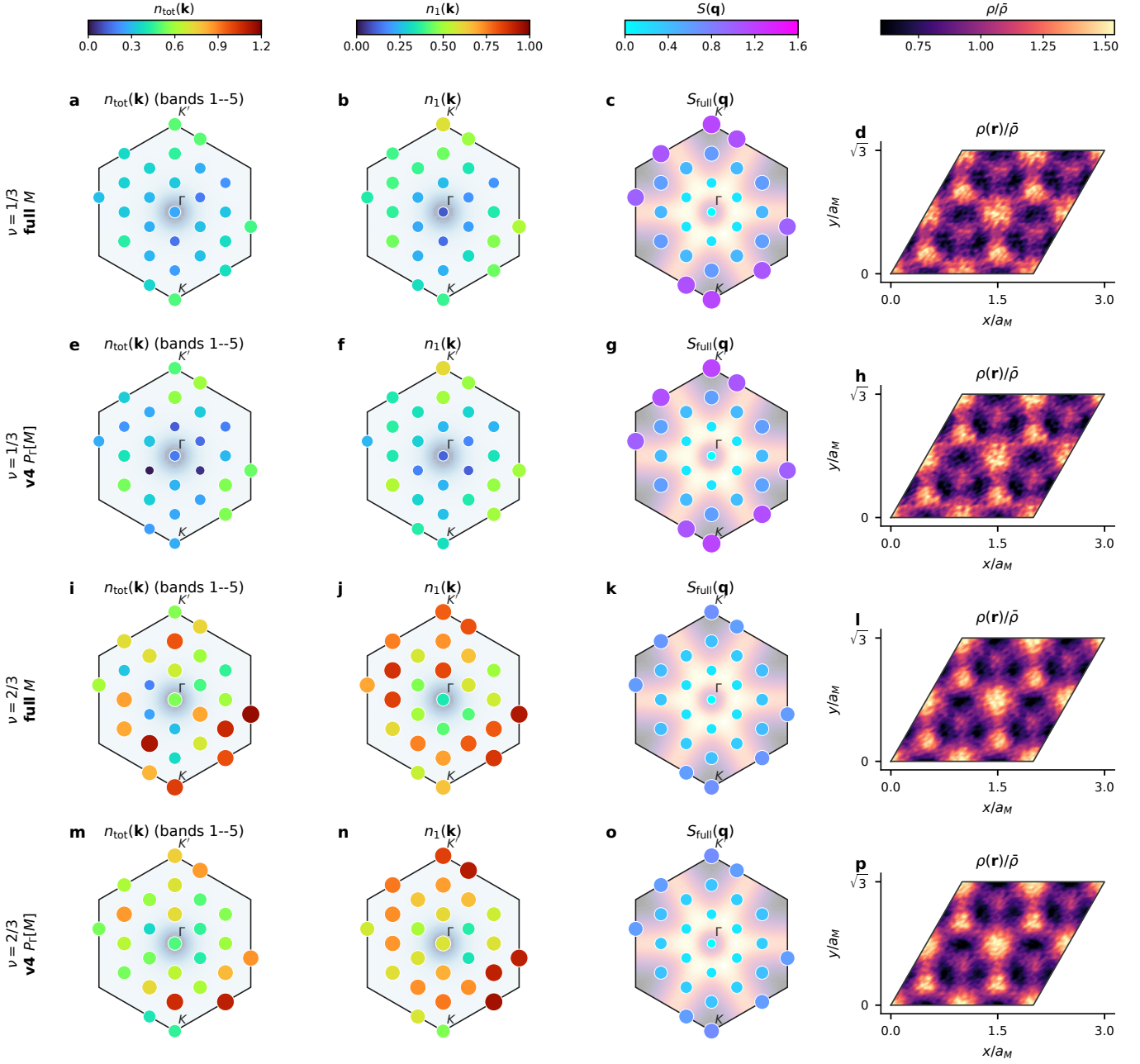


Figure 8: Twenty-seven-cell full- M and $P_T[M]$ observables. Rows compare both ansatzes at $\nu = 1/3$ and $2/3$. Columns show the Bloch-projected occupation summed over bands 1–5, the band-1 occupation conditionally normalized to N_e , the full static structure factor transformed from the real-space density-pair correlation, and folded charge density. Both momentum panels and the main $S(\mathbf{q})$ panel use exactly the 27 physical ED momentum vectors. A directly evaluated 37-vector, C_6 -closed grid is retained only for boundary rotation audits. No observable is C_3 averaged, and no ED momentum profile enters the first-band normalization. All values use the saved final 4,128-walker training ensembles without a separate validation chain.

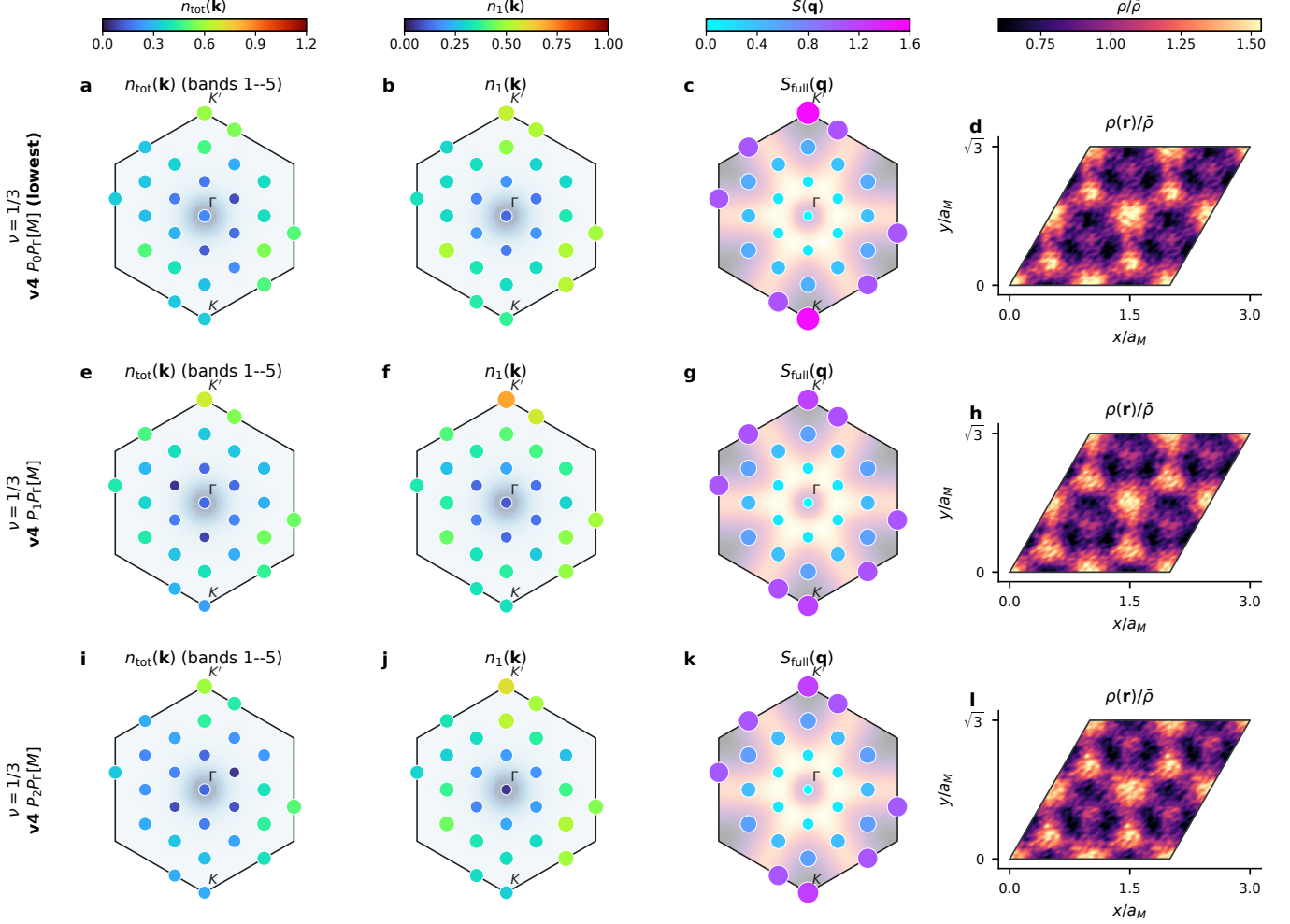


Figure 9: Twenty-seven-cell $\nu = 1/3$ outer-sector observables. Rows are the complete-wavefunction projectors P_0, P_1, P_2 ; the lowest final-ten branch is marked directly. Columns and all momentum-observable rendering conventions match Fig. 8 and the ED reference in Fig. 3. Shared color limits permit a direct visual comparison without sector-by-sector rescaling.

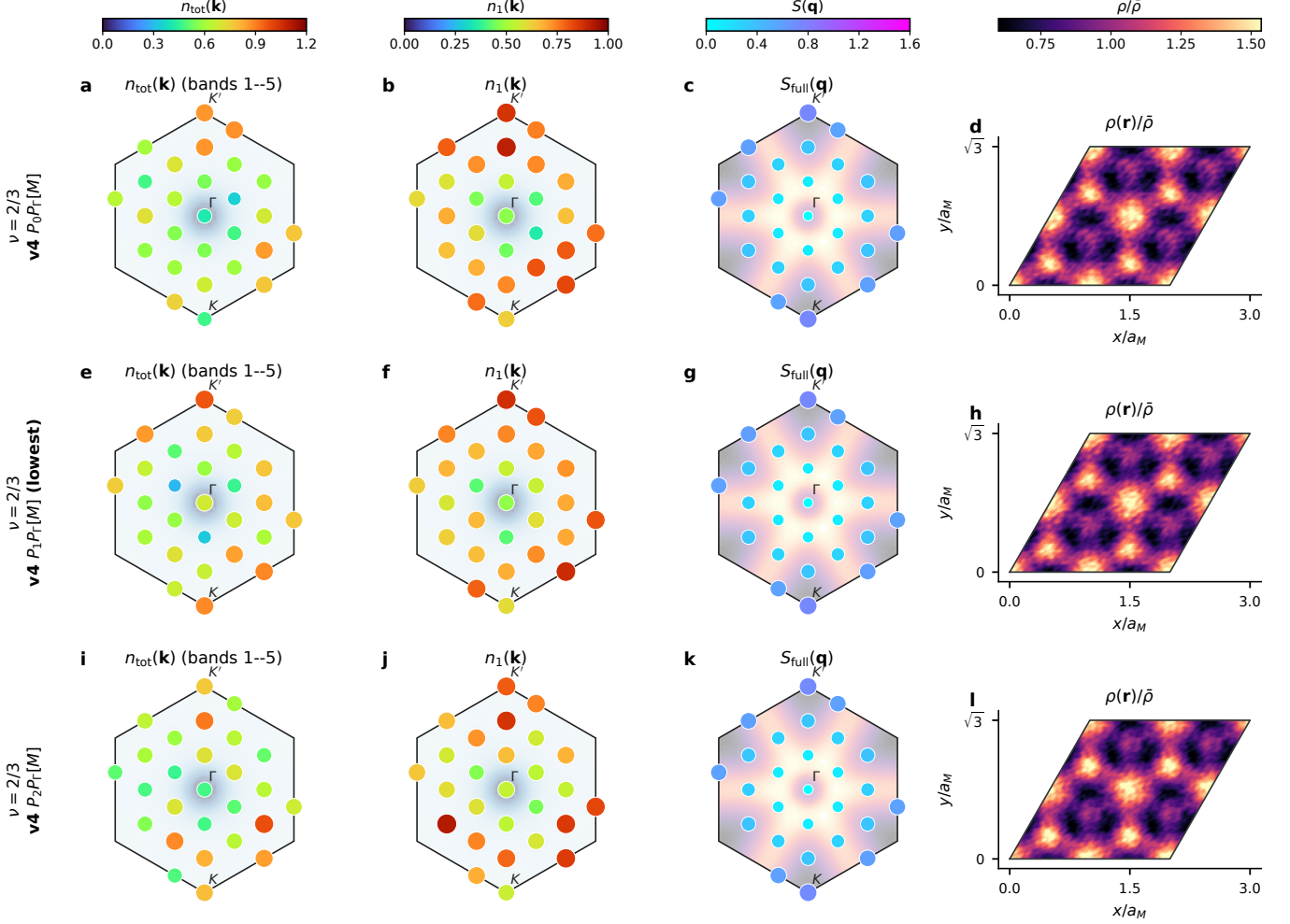


Figure 10: Twenty-seven-cell $\nu = 2/3$ outer-sector observables. Rows are P_0, P_1, P_2 and columns use the same definitions, color limits and marker rules as Fig. 9. The nearly degenerate final-ten $m = 1, 2$ energies are not by themselves evidence of a converged three-state manifold.

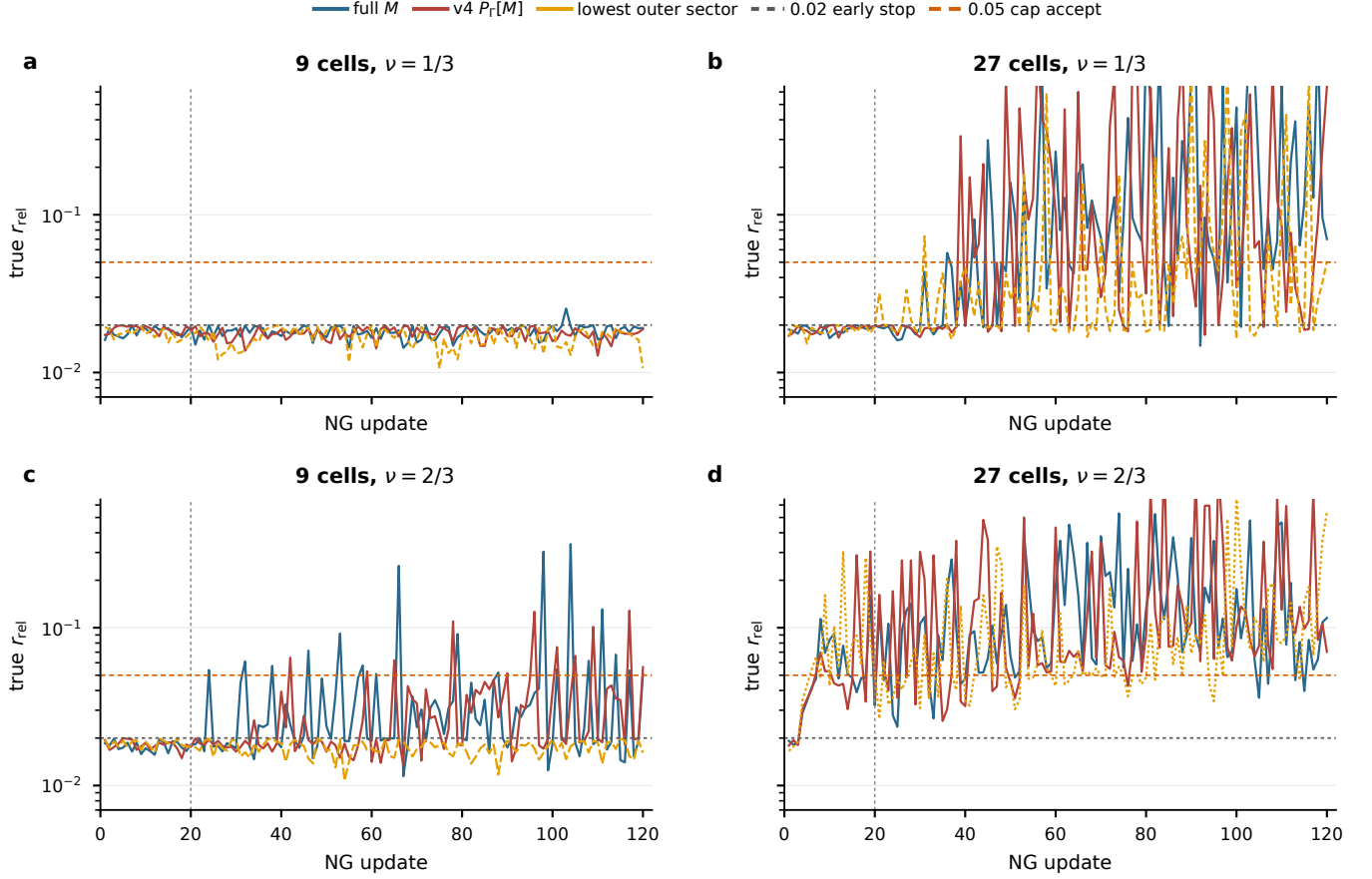


Figure 11: Supplementary nine/27-cell natural-gradient stability. True relative residuals are recomputed from $F - (S + \lambda I)\Delta\theta$. Columns compare nine and 27 cells and rows compare $\nu = 1/3$ and $2/3$. Horizontal lines mark the 0.02 early-stop and 0.05 cap-acceptance thresholds; the vertical line marks update 20. The 27-cell residual excursions and rejected updates show that the 120-step energies remain finite-optimization diagnostics.